Australian Government

Department of Employment and Workplace Relations

Office of the Chief Economist Data Visualisation Guidelines

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1. About these guidelines

Who: these guidelines are for staff in the economics branch of the Office of the Chief Economist (OCE) at DEWR. References to OCE in this document refer to the economics branch.

What: the guidelines relate mainly to static (not animated or interactive) explanatory charts. This guide complements the DEWR-wide brand guidelines and the Chief Data Officer's Data Style Guide.

When: if you're making a chart for a report, or presentation, or generally to show to someone else, you should follow these guidelines.

The guide makes it clear when you **must** do things a certain way, or when you merely **should** generally do things a certain way.

George Orwell's famous guide to writing concludes with the following: "Break any of these rules sooner than say anything outright barbarous." In that spirit, you should break any of the rules in this guide rather than make a visualisation that is inaccurate, inappropriate, or unattractive.

Why: Good, clear communication is vital to our work. If you do high-quality analysis but don't communicate the results in a way that your audience understands and remembers, your analysis is wasted. Communicating in a consistent, professional style instils confidence in our work and builds 'brand recognition' for DEWR and the Office of the Chief Economist (OCE). We have high standards, not only for analytical rigour but also for the clarity of our communication.

How: the guide is software-agnostic. The same principles apply whether you use R, Excel, PowerBI, or something else. Just as the APS Style Manual applies to all writing regardless of which word processor you use, these guidelines apply regardless of which software you use to make charts.

2. The basics of OCE charts

An OCE chart:

- **Doesn't mislead**: Care should be taken to ensure that your chart accurately represents the data and is unlikely to be misunderstood. See section 4, 'Don't mislead'.
- **Considers the audience**: The appropriate chart, message, and design differs for different audiences. See section 5, 'Consider your audience and context'.
- **Conveys a message**: Charts should have a key point, conveyed clearly using an active title, reinforced through colour and annotations. See section 6, 'Tell a story'.
- **Uses DEWR's colours:** Colours should be drawn from the DEWR palette. Combinations should be contrasting and accessible. See section 7, 'OCE style: use DEWR colours'.
- **Uses text thoughtfully**: All text should be in Aptos. Most text should be horizontal. Text should be at least 11pt for a document or 18pt for a presentation. Directly label series where possible. See section 8, 'OCE style: use text appropriately'.
- **Features a clean, minimal style**: Omit extraneous visual components and emphasise the data that conveys your key message. See section 9, 'OCE style: clean and minimal'.

3. Principles for OCE charts

Data visualisations **must accurately reflect the data, should tell a story,** and **must look professional and consistent.**[1]

Accuracy may seem obvious. No one generally sets out to make an inaccurate or misleading visualisation. But your charts can still accidentally mislead if you don't follow good visualisation practices. For example, secondary y-axes, truncated axes, and inappropriate colour schemes can all result in readers misperceiving your data.

Communication is about **storytelling**. Data visualisation is no different. Data does not speak for itself. Your visualisation should communicate a clear message. If it doesn't, your readers will take away their own message from the visualisation, and it may not be the one you intended. Or they may take away nothing at all.

Professional, consistent style is important. Visualisations should communicate a key analytical message, but also a meta message: this analysis was done by competent professionals who take pride in their work. Our style is focused and deliberate, not cluttered or messy.

Visualisations should be made with a specific audience in mind. Is your audience the Minister? Or colleagues who may understand the subject matter but not the data or method you've used? Or a teammate who has worked with you on the project? The choices you make for each of these audiences may differ, but the principles of good visualisation do not.

4. Don't mislead

No one in OCE will set out to make a misleading chart. But there are ways you can mislead readers unintentionally, through the choices you make in creating a visualisation. To avoid misleading readers:

- Don't use 3D
- Don't truncate your bar chart's y-axis
- Include the right amount of context
- Be careful with dual y-axes
- Include relevant caveats and data notes
- Ensure the data is consistent with your message

In addition to these specifics, always think through the way someone less familiar with your data, subject matter, analytical method, or chart type might perceive your visualisation. It can be useful to show your visualisation to a friend or colleague and see if their interpretation of the chart is the one you intended.

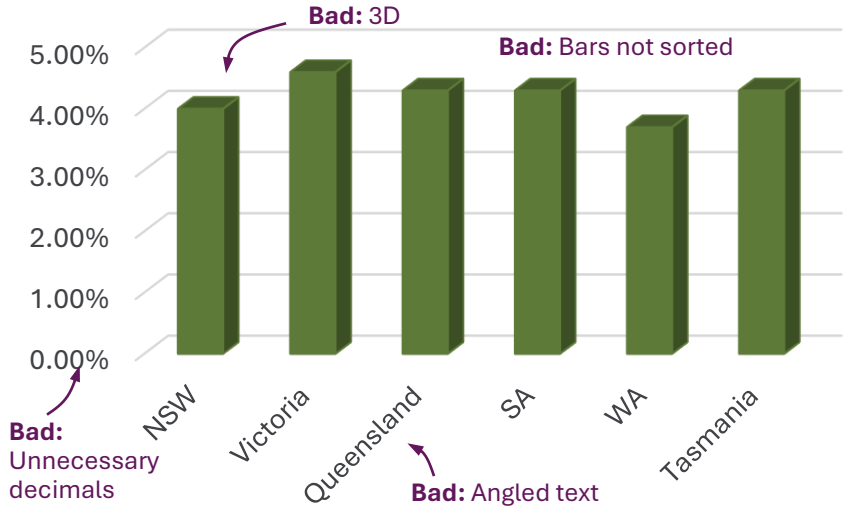
Don't use 3D

3D visualisation is a fad whose time has mostly passed. But occasionally you will still see 3D charts in the wild, like the one shown in Figure 1. The 3D effect makes it harder to read the bars' values and harder to distinguish between their lengths. It also adds unnecessary visual clutter. You **must not create 3D charts**.

[1] Based on principles set out in Wilke, C. 2019, *Fundamentals of Data Visualization,* O'Reilly Media. Available free online: https://clauswilke.com/dataviz/

Figure 1: 3D charts are bad, don't make them

Unemployment rate by state, Australia (July 2024)



Note: seasonally adjusted.
Source: ABS Labour Force

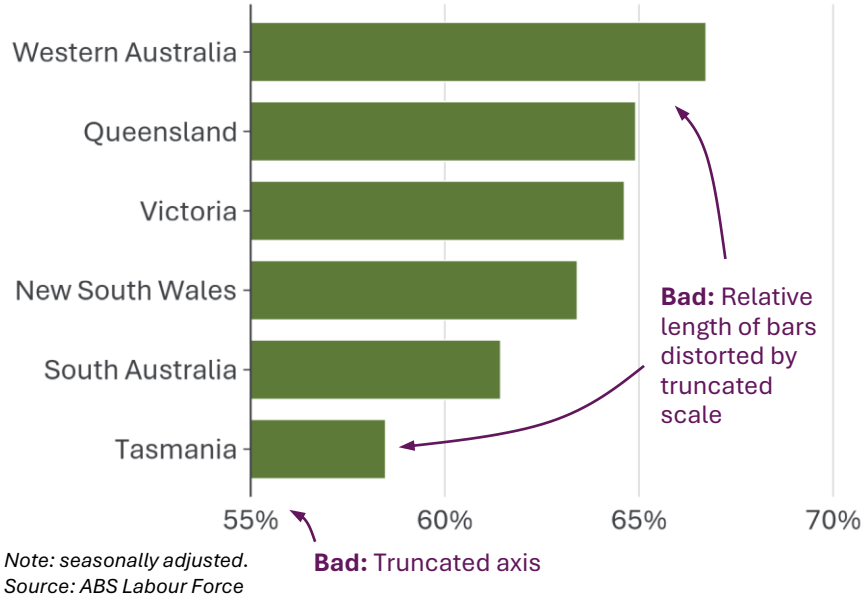
Occasionally you may be tempted to create a 3D chart with three variables, with one variable represented as depth on the chart. For example, you may want to show how a household's income varies with the hours worked of its primary and secondary earners. This is almost always a bad idea, as the result will be very difficult to read, even for experts who are familiar with the subject matter. Consider another approach instead, such as many small charts (known as 'facets' or 'small multiples').

Don't truncate your bar chart's axis

Some charts, such as bar charts[2] or area charts, use the length of a visual element to correspond to the data. This means that the data can be easily misperceived if you truncate the axis, as in Figure 2.[3] In Figure 2, WA's employment-to-population appears to be roughly triple that of Tasmania, whereas it's in fact only 1.08 times Tasmania's rate.

Figure 2: Bar charts with truncated axes mislead readers

Employment-to-population ratio by state, Australia (May 2025)



Note: seasonally adjusted.
Source: ABS Labour Force

This chart is misleading. You **must not truncate the axis of your bar or area chart**. The continuous axis on these charts must include zero.

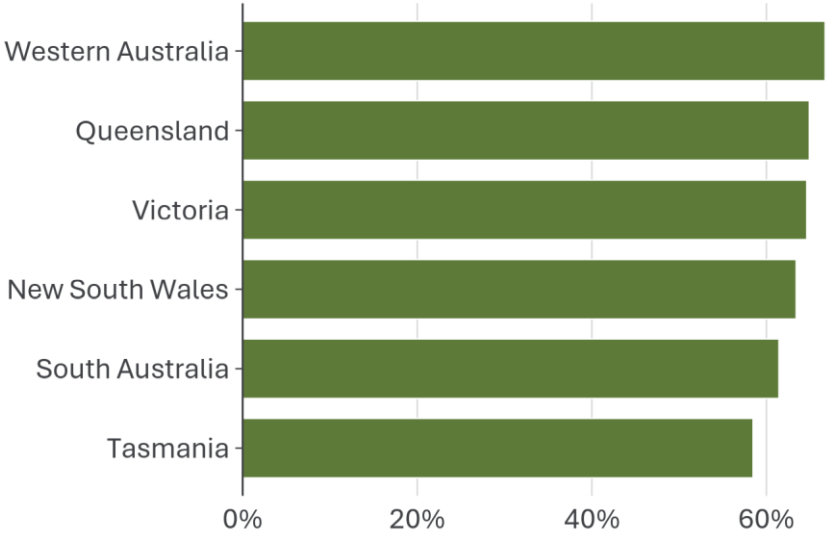
[2] Microsoft distinguishes between 'column charts' (vertical bars) and 'bar charts' (horizontal bars). It is otherwise conventional to refer to both as bar charts, a convention we follow in this guide.

[3] Almost all data visualisation guides state that the continuous axis for bar and area charts must always include zero. This is based on a range of research, for example Yang, B.W., et al. 2021, 'Truncating Bar Graphs Persistently Misleads Viewers', *Journal of Applied Research in Memory and Cognition,* 10(2):298-311.

You could fix Figure 2 by starting the continuous axis at zero, as in Figure 3. This chart is no longer *wrong* in the same way as Figure 2. But this could mislead in the other direction, by de-emphasising or disguising the variation that does exist between states. Figure 2 is a clear-cut case – do not make charts like this, bar charts' axis **must** always include zero. Figure 3 is a less clear-cut case and requires judgement: is a reader likely to accurately perceive your data and message from the chart? If not, consider another chart type that doesn't require the axis to start at zero.

Figure 3: An axis that starts at zero can mislead by hiding important variation in the data

Employment-to-population ratio by state, Australia (May 2025)



Note: seasonally adjusted.
Source: ABS Labour Force

As an alternative to Figure 2 or Figure 3, you could use some other chart type (such as a 'dot plot') that doesn't use length or area to encode data. Figure 4 provides an example of this – the truncated axis is acceptable for chart types like this.

Figure 4: Truncated axes can be OK for other chart types

Employment-to-population ratio by state, Australia (May 2025)

Note: seasonally adjusted.
Source: ABS Labour Force

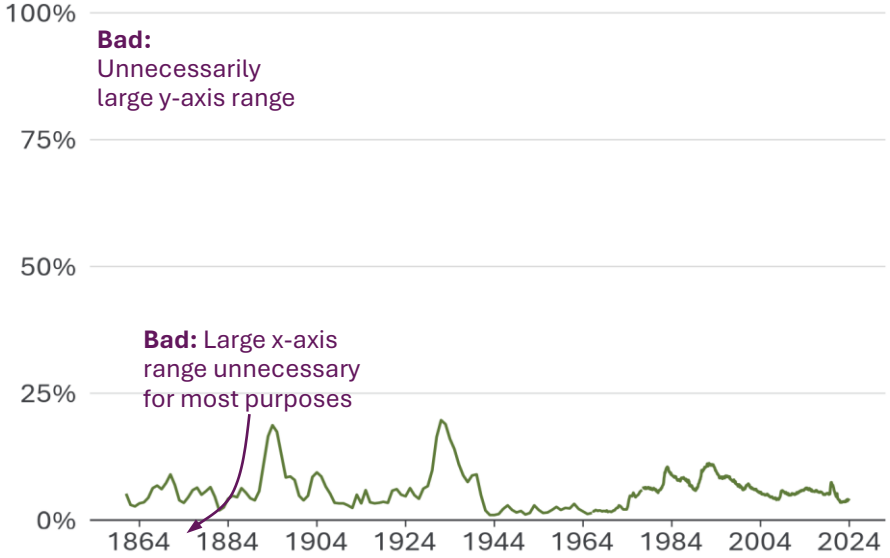
Include the right amount of context

The continuous axis on a bar or area chart should never be truncated, as in Figure 2. But it is not always necessary to include zero in the continuous axis for most other chart types, like line charts and dot plots. For these, the guidance is less definitive and more vague: **you should give viewers enough context** to understand your data.

It's possible to give too much context. Figure 5 is an example of this. The chart shows the unemployment rate, with the y-axis (vertical) extending from 0 to 100%. This is unnecessary, as the range extends far above and below the range observed in the data. Even the very high unemployment rates recorded in the Great Depression appear relatively modest on this chart. The y-axis provides too much context.

Figure 5: It's possible to provide too much context (bad)

Unemployment rate, Australia, per cent of labour force

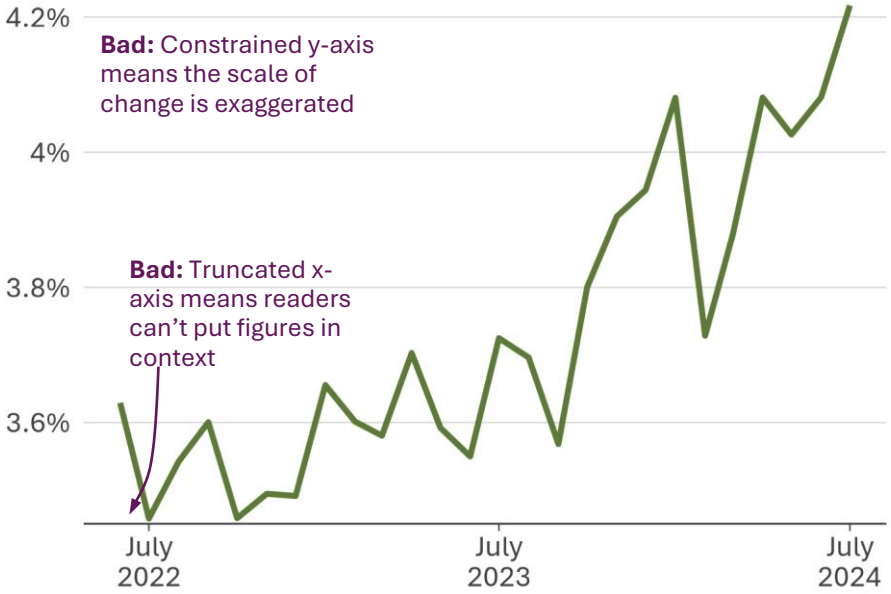


Note: seasonally adjusted monthly data from 1978; seasonally adjusted quarterly data from 1966-77; annual data prior to 1966.
Sources: ABS Labour Force; ABS Labour Force Historical Timeseries, Australia, 1966 to 1984; Butlin, Dixon and Lloyd 2014.

The x-axis (horizontal) in Figure 5 also provides more context than is required for most purposes. For example, if you were making a chart where the key message is "unemployment has started rising from its post-COVID lows" then a chart that includes data back to the 1860s provides too much context.

Figure 6: It's possible to provide too little context (bad)

Unemployment rate, Australia, per cent of labour force



Note: seasonally adjusted.
Source: ABS Labour Force.

Figure 6 shows the same data as Figure 5, but with aggressively truncated axes that do not give sufficient context. The constrained y-axis gives the impression of a very large change. The x-axis is also truncated, which means readers cannot put the current level of unemployment in context by comparing it to periods before mid-2022. The appropriate amount of context to show depends on your message and on the variation or comparison you are highlighting in the data.

Context can also be shown by including other relevant data (eg. other countries' unemployment rates). Judgement is required. Ask yourself: what would a reader, unfamiliar with this data, conclude from my chart? Include enough context so that this expected inference is the same as your intended message (see section 6, 'Tell a story').

Be careful with dual y-axes

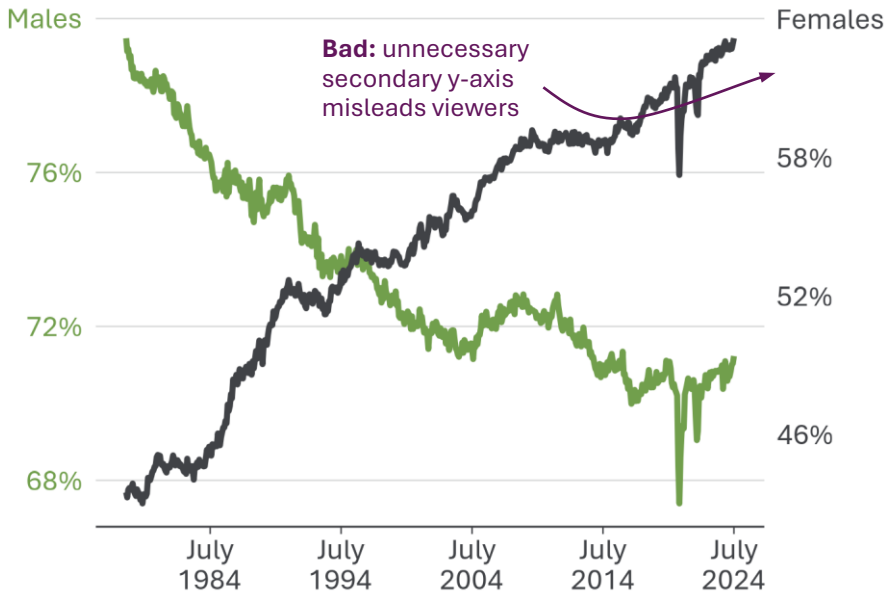
It is very easy to deceive viewers with dual y-axis charts[4], as in Figure 7. Here, the primary (left-hand) y-axis corresponds to the male participation rate, while the female rate is shown on the secondary (right-hand) axis.

Viewers of Figure 7 could easily be misled into thinking that female participation (the dark grey line) is higher than male (green). Although this is a particularly egregious example that is (hopefully!) unlikely to appear in a real-world publication, it is still easy to mislead with a dual y-axis chart.[5] It is easy to (intentionally or otherwise) exaggerate the degree of association between two variables in a dual-axis line chart.[6]

Most visualisation guides recommend against using dual y-axis charts in most, or sometimes all, cases. You **should consider alternatives to dual-axis charts.** If you can show the series on the same axis, do that. Otherwise, consider using small multiples, converting to index numbers, using a scatterplot, or showing a ratio between the series – see 'Example 3: Dual-axis line charts' (p.30) for examples. In *rare* cases a chart with a secondary y-axis may be the most appropriate choice, but ensure you've considered the alternatives.[7] If you do create a dual-axis chart, you **must ensure the axes are clearly labelled, with ranges that reduce the potential for misunderstanding.** Consider colouring the axes to align with the colour of the series.

Figure 7: Secondary axes like this can easily mislead

Labour force participation rate by sex (Australia)



Note: seasonally adjusted.
Source: ABS Labour Force.

Dual y-axis charts are sometimes used to overlay two charts – for example a chart that shows the unemployment rate on one axis and employment growth on the other. While the potential for these to mislead is low, they are generally unattractive, difficult to read, and would work better as two charts.

[4] This refers to charts with y-axes that differ from one another. Charts with a secondary y-axis identical to the primary axis are not misleading, though inconsistent with OCE style. Charts with a secondary axis that is a transformation of the primary axis are also not misleading (for example, a secondary axis showing Fahrenheit and a primary axis showing Celsius).

[5] See, for example, Isenberg, P., et al (2011), 'A study on dual-scale data charts', *IEEE Trans Vis Comput Graph,* 17(12):2469-78.

[6] See Healy, K. 2016, 'Two Y-Axes', *Kieran Healy Blog*. Available online: https://kieranhealy.org/blog/archives/2016/01/16/two-y-axes/

[7] See Ellis, P. 2016., 'Dual axis time series plots may be ok sometimes after all', *Free Range Statistics*. Available online: https://freerangestats.info/blog/2016/08/18/dualaxes.

Include relevant caveats and data notes

Your chart should lead with your key message (see section 6). **You should not lead with or emphasise the caveats and notes about your data** except in the rare case that the uncertainty or caveats are the point of the visualisation. But, although caveats should almost never be the main point of the chart, these caveats and notes must always be included.

Your notes **must indicate whether the data was seasonally adjusted** or otherwise smoothed. It **should note if there are important omissions from the data**, quirks that the reader should be aware of, or if observations are missing or have been removed.

You **must always list the source**. Don't just say "ABS", be more specific, such as "ABS Labour Force". Include source information under each chart – even in cases where multiple charts in a document have the same source, each chart should have this information underneath.

The notes and source information do not need to be part of the chart image itself. These can be below the chart in the document or presentation in which the chart is being used – just as in this document.

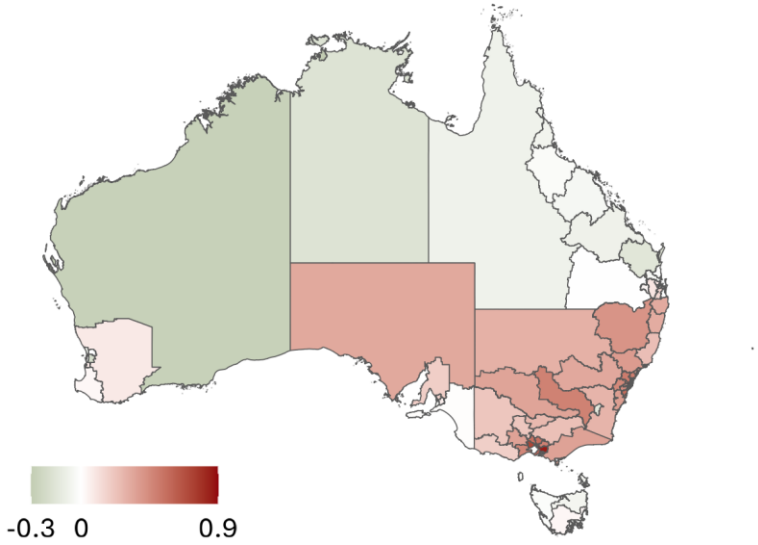
Ensure the data is consistent with your message

Your charts should contain a clear message conveyed in an active title (section 6). It's important that this message aligns with the data. The data should support your message, not contradict it.

An example of a chart where the data does not align with the message is Figure 8. This chart shows that most of the *land mass* of Australia experienced a fall in the unemployment rate over the year to January 2025. But it's not accurate to say – as the title does – that unemployment fell in most regions. Unemployment rose in many geographically small regions. This is less apparent on a choropleth map like this.

Figure 8: 'Unemployment fell in most regions of Australia in the past year' would be a misleading title for this map

Change in unemployment rate by SA4 region, year to January 2025, percentage points



Note: Difference in 12 month rolling average of ABS modelled estimates.
Source: ABS Labour Force (detailed).

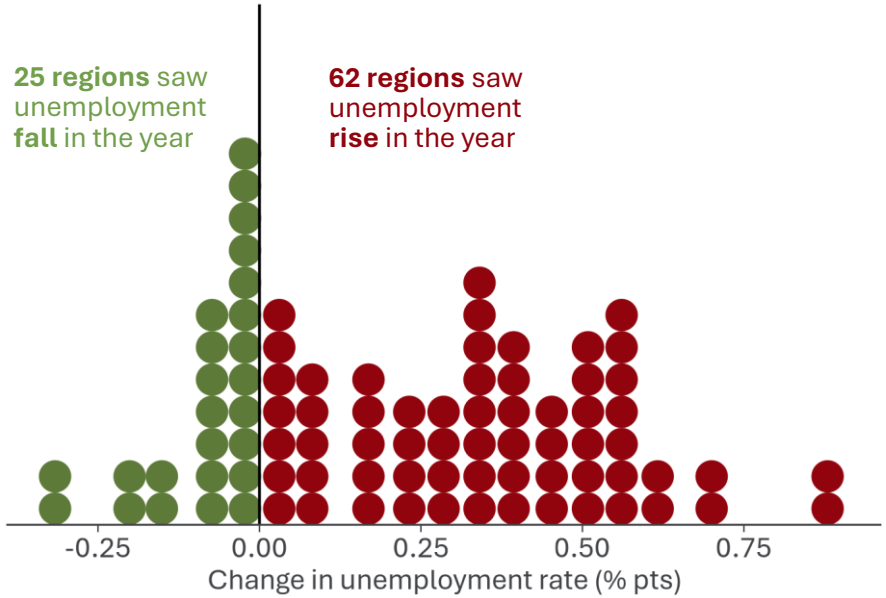
There are many ways that the visualisation and/or the title could change to bring them into alignment. Figure 9 shows one option, using the same data as Figure 8. The title has changed to better reflect the data. Rather than use a map, Figure 9 shows the distribution of changes in unemployment rates across SA4 regions, which makes it clear that more regions experienced a rise than a fall in unemployment.

The right approach to visualising the data, and the right message to extract from the data, will depend on the circumstances. But you **must always ensure that the main message of your chart is consistent with**

the underlying data to ensure you don't mislead readers and your audience takes away the message you intend.

Figure 9: Unemployment rose in most parts of Australia

Change in unemployment rate by SA4 region, year to January 2025, percentage points

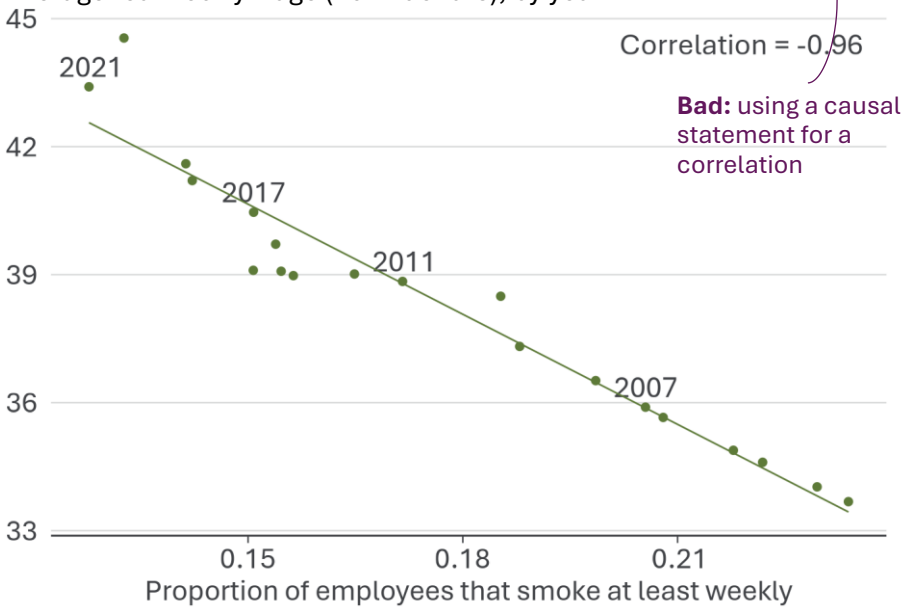


Note: Difference in 12 month rolling average of ABS modelled estimates.
Source: ABS Labour Force (detailed).

Figure 8 is a clear example of an inaccurate chart, where the message does not align with the data. There are many other ways to misrepresent your data that you should be careful to avoid. One way to mislead is by representing correlations or associations as causal relationships, as in Figure 10. Here, a coincidental association between two time series is described as causal, which is misleading.

Figure 10: It would be misleading to say 'more smoking causes lower average wages'

Average real hourly wage (2021 dollars), by year



Source: HILDA.

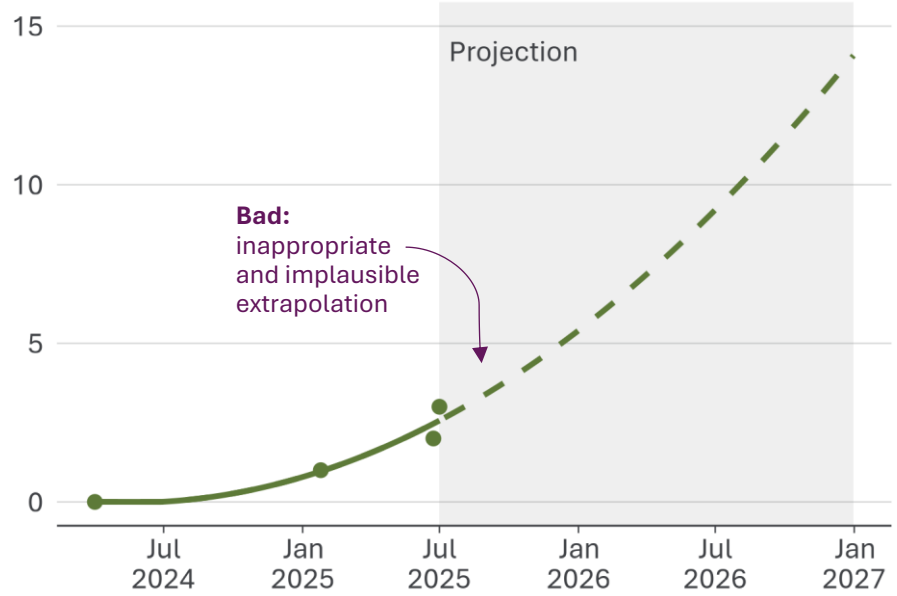
The bar for making causal claims is very high. To make such a claim, it needs to be theoretically defensible and also justified through evidence that transcends mere correlation. Generally, causal claims need to be supported by evidence either from an experiment (like a randomised controlled trial) or a natural experiment, examined through techniques such as difference-in-differences.

Research design and causal inference is beyond the scope of this data visualisation guide[8], but you should always be wary of the potential that readers might interpret associations in your data in causal terms.

Your charts can also mislead by extrapolating data, or otherwise inferring a strong relationship between variables that isn't there. For example, Figure 11 projects the number of branch heads in OCE, based on the implausible assumption that current trends will continue.

Figure 11: Extrapolating your data can be misleading

Number of branch heads in OCE



Source: Internal DEWR data.

[8] A good overview of the toolkit of modern applied economics for uncovering causal relationships is provided in Cunningham, S. 2021, *Causal Inference: The Mixtape,* Yale University Press. Available as a free online text: https://mixtape.scunning.com/. See also Huntington-Klein, N. 2025, *The Effect: C An Introduction to Research Design and Causality (Second Edition),* Chapman & Hall. Available as a free online text: https://theeffectbook.net/

5. Consider your audience and context

Appropriate communication styles vary with the audience. If you're explaining some analysis you've undertaken, your message, style and level of detail should differ if you're explaining it to an academic, or a Minister, or a colleague in a policy team, or to a member of the public.

It's vital that you take your audience into account when designing your chart. This is expected of us as public servants; the Integrated Leadership System includes "seeks to understand the audience and tailors communication style and message accordingly" in the profile of every level from APS3 up. Good communication is not optional.

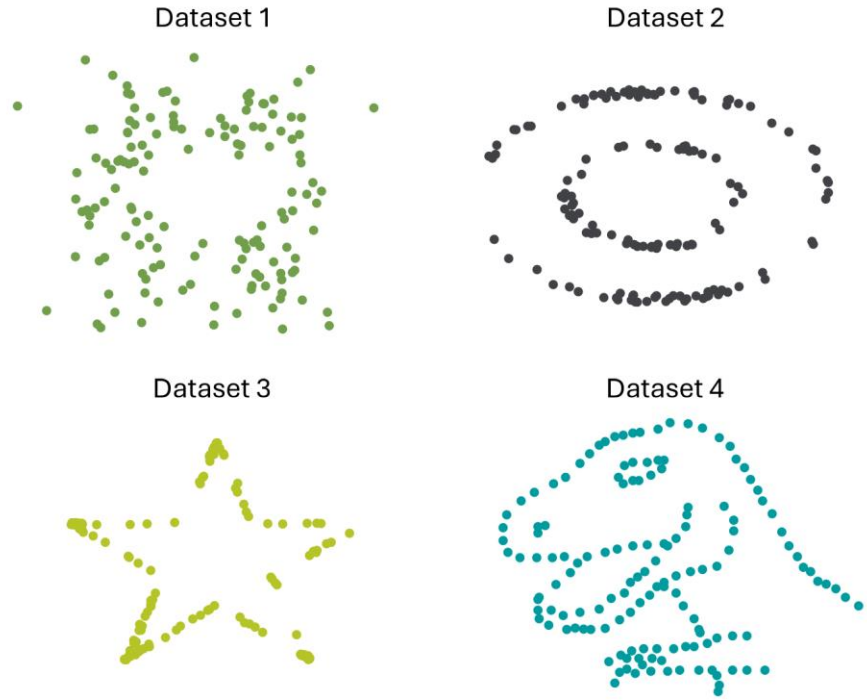
Explore your data, but don't use exploratory visualisations for explanation

An excellent data visualisation textbook begins with the sentence: "You should look at your data."[9] This is because it's easy to be deceived by your data if you don't visualise it.

Figure 12 provides a cautionary tale about what can be hidden if you don't visualise your data in the exploration stage. **Each of the datasets depicted in Figure 12 has the same standard deviation and mean** on both the vertical and horizontal axes. If your exploration of this data was limited to calculating summary statistics, you would miss the substantial difference between these datasets. Exploratory visualisations should be a key part of your analysis process.

Figure 12: Look at your data, because summary statistics can hide surprises

Four datasets with identical means and standard deviations on both axes



Source: 'Datasaurus dozen' by Alberto Cairo.

When you're exploring a dataset, you should make many charts to try to understand the data and explore topics of interest. Because these charts are at the exploratory stage, they will often contain a lot of detail, will probably not be focused on making a key point, and will not have a refined and consistent style.

[9] Healy, K. 2018, *Data Visualization: A Practical Introduction,* Princeton University Press. Available free online: https://socviz.co/

Figure 13 is an example. This chart is fine for exploratory purposes, to help an analyst and perhaps their close collaborators to understand the data. Figure 13 would be terrible as an explanatory visualisation. It's cluttered, unfocused, unrefined, and does not make a clear point.

It's important to recognise that charts like Figure 13 may work perfectly well when the audience is you, the person analysing the data, but they work terribly when the audience is anyone else. A common mistake is to simply take exploratory visualisations, perhaps style them slightly, and include them in a document or presentation.

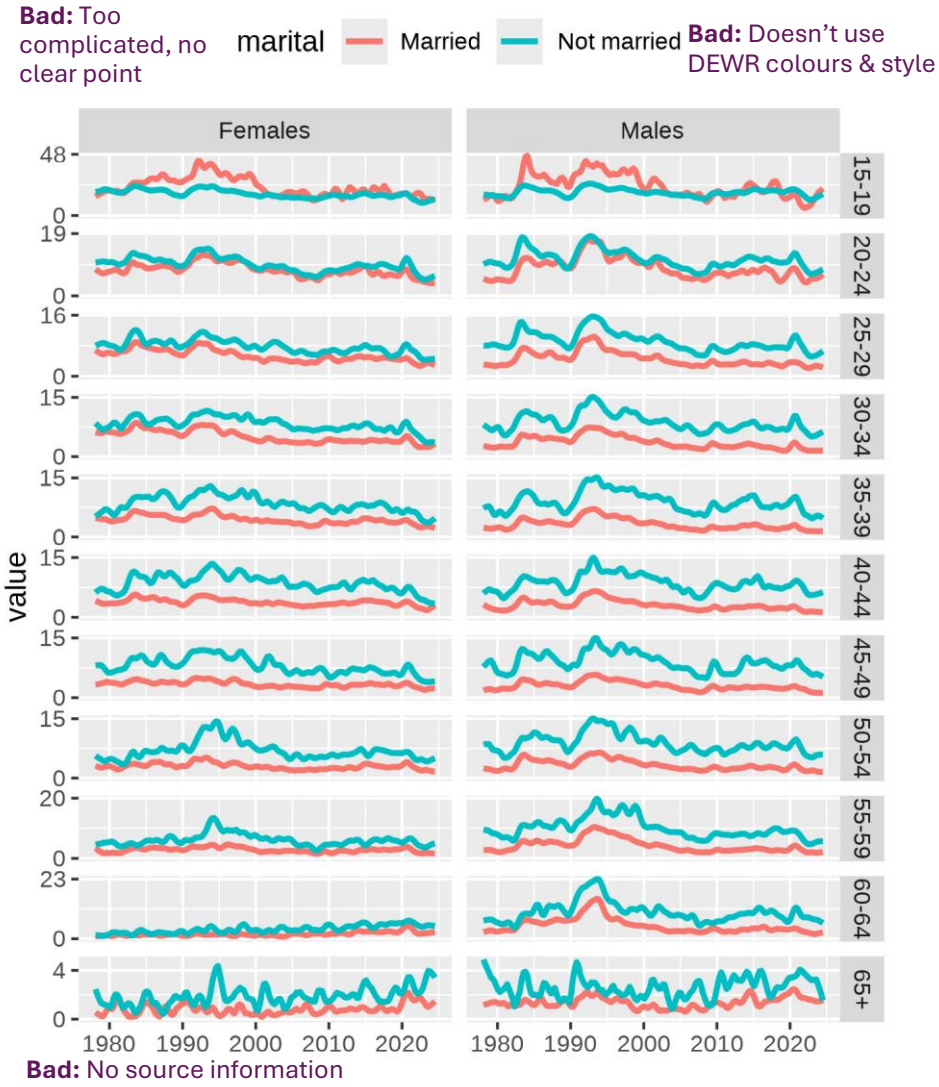
Your approach must be different when your audience is not you or your close collaborators. **You must not use exploratory visualisations for explanation.** Instead, focus on what you feel is the key message of the data. Strip out detail until the chart clearly conveys this message (see section 6 for more on this).

Think carefully about your audience

Your chart draws on some data, generally applies some analytical method (perhaps as simple as a mean or a sum), and conveys a message. Your audience may be unfamiliar with the data, method, and/or subject matter. These may therefore require explanation. For an audience of economists, you will not need to explain techniques such as linear regression – for some audiences, even something as simple as linear regression may require careful, jargon-free explanation. Some audiences may be familiar with your data and method, but unfamiliar with the subject matter.

Figure 13: A chart that may work for exploration is usually bad for explanation

Unemployment rate by age, sex, and marital status (per cent)



Different audiences will also have a different level of interest in aspects of your work. An audience of economists may be very interested in your identification strategy, or in the details of your statistical analysis. Policy colleagues may be most interested in the high-level findings from your work and their policy implications.

Think carefully about who your audience is. Think about what they already know, and what they most need to know from your analysis. Your **visualisations should be tailored to your audience**. What is appropriate for one audience may not be appropriate for another.

Your audience almost certainly consists of smart, capable professionals who understand many things very well. But they may not be familiar with all aspects of what you are presenting. Tailor your language – particularly jargon – to the audience. Choose chart types that your audience is likely to be familiar with. Aim for clarity, not needless complexity.[10]

Don't assume the same visualisation will work for documents and presentations

A common visualisation mistake is to take charts that have been produced for a written document, copy-paste them onto a PowerPoint slide, and present it to an audience. This usually works poorly. Presentations are visual aides used to complement a speaker's words. It is a fundamentally different format to a written document that is intended to be read by a person at their desk.

Complicated charts can sometimes work in a written document, depending on the audience. They rarely work in presentations, unless the presenter takes care to build up to and explain them. Charts in presentations should be even more focused on making a single key point as opposed to conveying multiple aspects of the data. Charts in presentations should feature much larger text than those in written documents- see section 8, 'OCE style: use text appropriately'.

Think about your product as a whole

It's rare that you will produce a document or presentation that includes only one data visualisation. There is usually a suite of visualisations that form a whole. There are two important things to consider: first, make sure that you vary your chart types within the document or presentation. Second, make sure you have consistency of style and colours.

You want your audience to remember your messages. This is less likely to happen if you subject them to an endless parade of charts that look the same. A long sequence of similar-looking bar charts will blur into one another –your audience will be less likely to comprehend and retain the message of each separate chart. This is not a reason to use wacky, novel charts for the sake of it, but it is a reason to carefully consider whether at least some of your charts could be presented differently.

Having too many similar charts can be a bad thing, but consistency across charts is important. This consistency should include the consistent use of colours across visualisations. If the 'eucalyptus' colour is used for New Zealand in one chart, don't use 'plum' for New Zealand in the following chart. Consistency of visual language across charts within your document or presentation reduces the cognitive load for your audience – they will be more able to quickly grasp what your data is showing and the message you're conveying.

[10] See section 29.2, 'Make a figure for the generals', in in Wilke, C. 2019, *Fundamentals of Data Visualization,* O'Reilly Press.

6. Tell a story

When you make a chart, you are trying to convey a message to your reader or audience in a way that they will understand and remember. Your visualisation may convey more than one point, but **charts should usually have one main point**. The choices you make – from the chart type, to what you emphasise, to how to summarise your message in an active title – should reflect and convey this main point. Try to ensure the message readers take from your chart is the one you intended to convey.

As Claus Wilke puts it in *Fundamentals of Data Visualization:*

> *any communication creates a story in the audience's minds. If we don't provide a clear story ourselves, then our audience will make one up. In the best-case scenario, the story they make up is reasonably close to our own view of the material presented. However, it can be and often is much worse. The made-up story could be "this is boring," "the author is wrong," or "the author is incompetent."*[11]

If your chart doesn't have a point, consider whether you need a chart at all. What purpose is it serving, if it is not conveying some message about your data that you feel is important for readers to understand?

The story you are telling with your visualisations should be based in your data. It should be of interest to your readers or audience. It should never be "I am a very smart person who has done complicated analysis and you should be very impressed," which is the implicit message too many economists and analysts try to convey. Your message should also never be "our data is very rich and detailed and complicated."

Once you have identified your main message, think about whether your chart can be simplified to convey it more clearly. Removing extraneous detail can often help with this. Never include extra detail just for the sake of it, or at the expense of clearly conveying your main message.

Use active titles to convey a message

Your chart should have a point. Your chart's title should summarise that point in a short, readable sentence.

An OCE chart should almost always have an 'active title' that describes, in a short sentence, the key message of the chart. Readers should be able to glean the main point of the visualisation from the title. **Charts in reports should generally 'stand alone'** – they should be capable of being understood when read in isolation. A reader should be able to skim your report, read the chart title, and take away the message.

Table 1 gives some examples of active titles, along with the subtitles that could accompany them. **Do not use a subtitle in place of a subtitle.** For example, "unemployment rates by state" would work as a subtitle but not as a title. A title may be "unemployment is rising everywhere except WA".

OCE charts should have active titles in almost all cases. However, there may be *occasional* instances where they are inappropriate. For example, in *some* straight factual summaries of information you may omit them, such as on simple factsheets. It may also be impractical to include active titles in interactive, dynamic dashboards, as the appropriate active title may change when the data changes. In these cases, the title should be omitted, with the subtitle remaining. If in doubt, include an active title.

---

[11] See Chapter 29, 'Telling a story and making a point' in Wilke, C. 2019, *Fundamentals of Data Visualization,* O'Reilly Press.

Table 1: Most charts should have active titles

Examples of active titles and accompanying subtitles

Active titles	Accompanying subtitles
Unemployment is rising in every state except WA	Unemployment rates by state (per cent)
Youth unemployment has risen, but remains below pre-COVID levels	Youth unemployment rate, Australia (per cent)
Most people who have a job 26 weeks after leaving services still have a job 2 years later	Average fortnightly payroll earnings: 26 weeks and 2 years after anchor date (dollars per fortnight)
People with more education tend to earn more	Average earnings by level of highest educational attainment, 2023 (dollars per week)

Use pre-attentive attributes to convey a message

Jon Schwabish, in 'An Economist's Guide to Visualizing Data'[12], asks readers to count the occurrences of the number 3 in the following set:

1269548523612356987458245
0124036985702069568312781
2439862012478136982173256

And then repeat the task with this set of numbers:

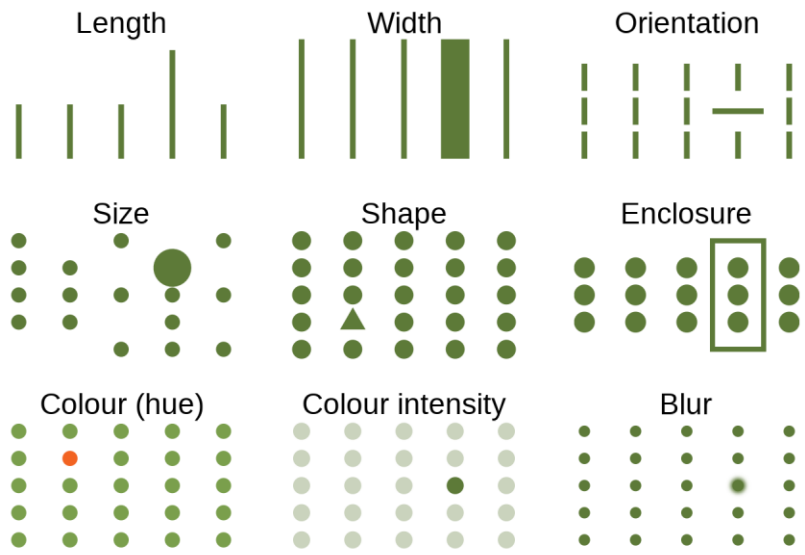
1269548523612356987458245
0124036985702069568312781
2439862012478136982173256

It is easier to find the 3s in the second set, because they are bolder and darker than the other numbers. Boldness (line width) and darkness (colour intensity) are *pre-attentive attributes* – aspects of your chart that will jump out at readers and be immediately apparent before they have focused their attention.[13] Figure 14 shows some pre-attentive attributes.

Your visualisations should carefully use these attributes to draw attention to the aspects of your data that are most relevant for your main message. They support and reinforce your main message, conveyed in the active title. Don't use pre-attentive attributes arbitrarily. Using these attributes draws readers' attention to something.

Figure 14: Pre-attentive attributes catch readers' attention

Common visual elements that the brain processes quickly



[12] Schwabish, J., 2014, 'An Economist's Guide to Visualizing Data', *Journal of Economic Perspectives,* 28(1), pp. 209-234. Available online: https://www.aeaweb.org/articles?id=10.1257/jep.28.1.209

[13] Healey, H and Enns, J, 2012, 'Attention and Visual Memory in Visualization and Computer Graphics', *IEEE Transactions on Visualization and Computer Graphics,* 18(1), pp.1170-1188. Available online: https://ieeexplore.ieee.org/document/5963660/

Put your numbers in context

Numbers are useless without context. If I tell you your blood pressure is 125/85, these numbers will not mean anything unless you have some pre-existing knowledge of what a normal, safe reading might be. Similarly, if I tell you "the female labour force participation rate was 67.0 per cent in May 2025" this number doesn't mean much without further context. Putting your numbers in context is a key part of conveying a message.

These are all true stories:

- The female labour force participation rate is substantially higher than it was a generation ago
- Female labour force participation in May 2025 was slightly below where it was in January 2025
- Female participation is substantially higher than it was before COVID-19
- The female participation rate is below the male rate
- The gap between the male and female participation rates has closed substantially in the past several decades
- Australia's female participation rate is above the OECD average and above many comparable countries
- Australia's female participation rate lags behind the highest rates among advanced economies
- The female participation rate no longer drops sharply for women in their late 20s and early 30s, as it used to
- There is still a substantial 'motherhood penalty' in Australia, due to lower participation among other effects.[14]

The 67.0 per cent female participation rate does not convey much meaning by itself. It needs to be put in context through comparison to other numbers – the female participation rate at various points in the past, or in other countries, or the male rate. All of these comparisons tell a different story. You need to decide which story you are telling.

Almost all data visualisations essentially involve comparing numbers to one another – the choice of which numbers to show and compare is a crucial one and shapes the message viewers will take from your chart.

Once you are clear on the message you are trying to convey with your chart, ensure that the comparison you are making in the chart is an appropriate one. Does it reflect and support your message? Is it accurate and not misleading?

[14] Bahar E, Bradshaw N, Deutscher N and Montaigne, M (2023) 'Children and the gender earnings gap: Evidence for Australia', Treasury Working Paper 2023-02, The Treasury.

7. OCE style: use DEWR colours

Consistent colours help to build a brand. Using a standard palette ensures that our charts will look harmonious when placed alongside each other in a document or presentation and ensures that readers can immediately identify them as DEWR OCE charts.

Sticking to a standard palette also takes the guesswork out of chart-making : you don't need to choose colours, they're already chosen for you. Using pre-chosen colour combinations also means your chart will be accessible to people with vision impairments, including colour blindness (colour vision deficiency).

Colour leaps out at readers – it's a 'pre-attentive attribute' of your chart (see section 6). Don't use colour arbitrarily, use it to highlight important elements or differences in your data.

Always use the DEWR palette

DEWR has an official colour palette, set out in the Brand Guidelines and shown at Figure 15. **Your visualisations must use the colours from the official palette (Figure 15)**. Colours from the primary palette should generally be favoured over the secondary palette, which should be favoured over the tertiary palette.

Choose a colour combination that maximises readability by minimising the similarity between colours. You must be conscious of how your colour combination may appear to people with visual impairments.

Figure 15: Data visualisations must use these colours

Official DEWR palette

Primary palette

Graphite #404246	Eucalyptus #719F4C	Dark Eucalyptus #5D7A38	Grey #A4A7A9
Mid grey #D7D8D8			

Secondary palette

Plum #62165C	Lime #B5C427	Teal #009B9F	Navy #0D2C6C

Tertiary palette

Blue #287DB2	Red #91040D	Cobalt #004F9D	Purple #573393
Yellow #E9A913	Pink #D37190	Mint #47BFAF	Orange #F26322

Generally use OCE's default colour combinations

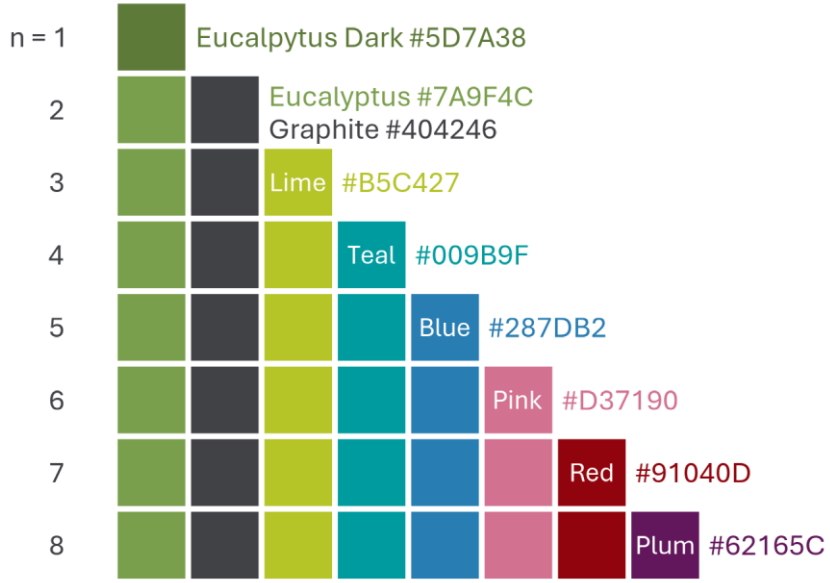
Your chart must use the DEWR palette from Figure 15. But you can choose which colours from the palette you use, subject to some constraints: **your colour combinations must work well together, have sufficient contrast and be accessible**.

OCE has default colour combinations, shown in Figure 16. For example, a two-colour chart should generally use Eucalyptus and Graphite, and so on. **You should generally use the colour combinations in Figure 16,** but you don't have to. Alternatives are acceptable – for example, a single-colour chart may use Eucalyptus Dark (the OCE default), Eucalyptus, or Graphite. Just make sure your chosen colours work well together and have high enough contrast.

Figure 16: The default OCE colour combinations

For charts with 'n' colours, these combinations should generally be used:

For charts with 'n' colours, these combinations should be used:



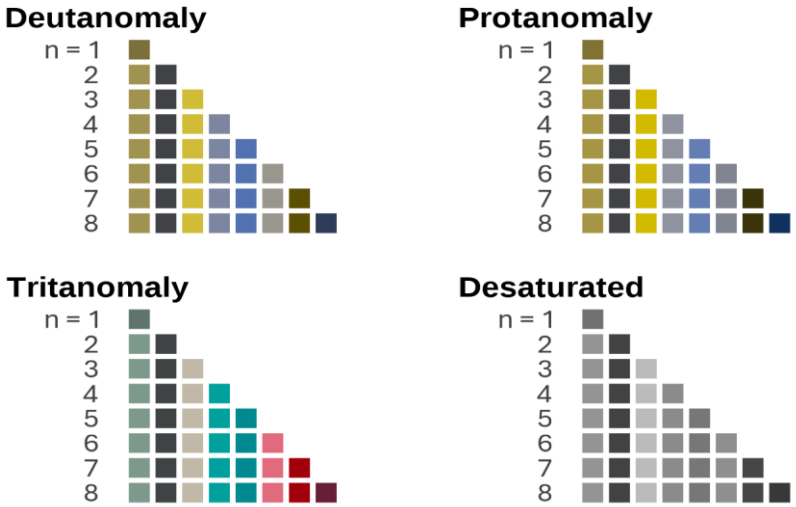
You should not generally use more than three or four colours in a chart. In most cases, you should consider reducing the number of colours. An example of a chart with too many colours, and potential approaches to reducing the number, is shown in the 'Example 1: Spaghetti charts' section. Gray is the most important colour in data visualisation.[15] Use light or mid grey to de-emphasise data, indicate missing data, to give context, or to annotate your chart. **Do not use light or mid grey like other colours.**

Use colour combinations that are accessible and don't reinforce stereotypes

Whatever colour combination you use, make sure there is a sufficiently high degree of contrast between the colours. Ensure that the difference between the colours is apparent to people with vision impairments. The default OCE colour combinations have been tested to maximise their readability, including for people with common vision impairments. Figure 17 shows how the OCE default combinations appear to people with common forms of colour blindness.

Figure 17: Colour combinations must be accessible

OCE combinations as viewed by people with common vision impairments



Don't use 'traffic light' colours, as red and green are not easily distinguished by people with common vision impairments.

[15] Muth, L. C. 2022, 'A detailed guide to colors in data vis style guides', *Datawrapper Blog*. https://blog.datawrapper.de/colors-for-data-vis-style-guides/

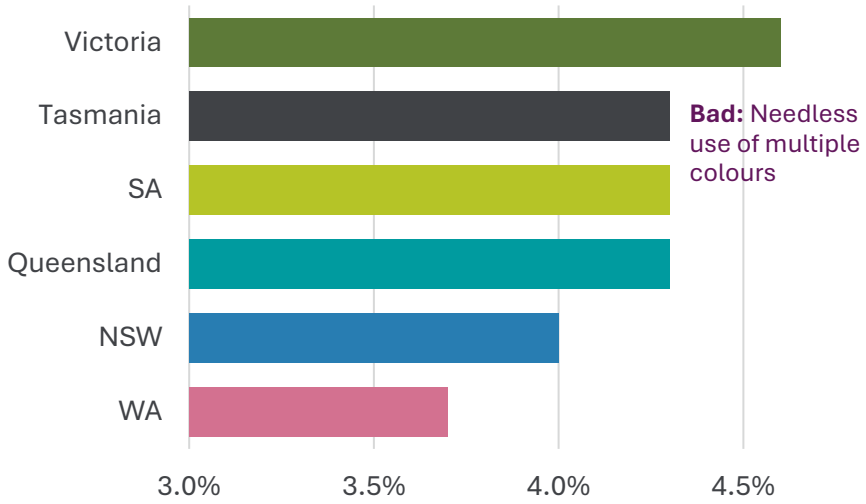
Be careful with colours that have strong cultural associations or that reinforce stereotypes, such as pink for women and blue for men. In fact, generally don't use pink and blue together in a two-colour chart even if the data don't have anything to do with women and men, as the colours have a strong association with those categories.

Use colour deliberately, sparingly, and consistently

You generally shouldn't encode the same information in multiple ways. For example, Figure 18 shows a bar chart where colour is used unnecessarily. States are already distinguished from on another by their position on the vertical axis – the colour adds no additional information.

Figure 18: It's generally a bad idea to encode information in multiple ways, like position and colour

Unemployment rate by state, Australia (July 2024)

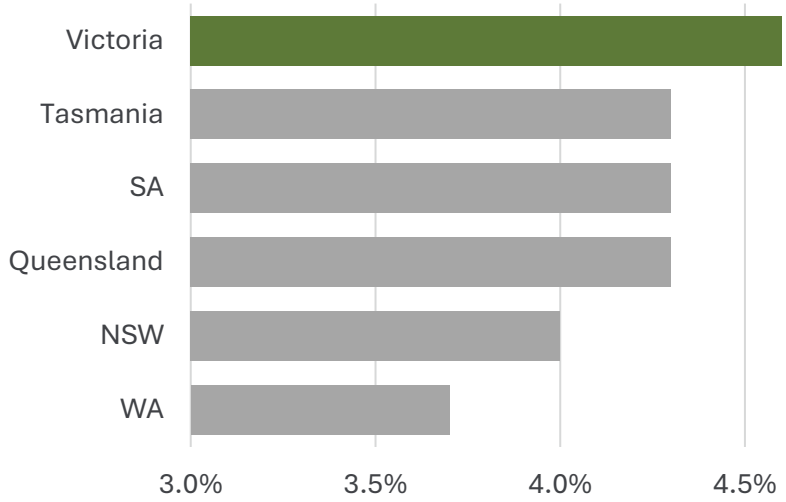


Note: seasonally adjusted.
Source: ABS Labour Force

Colour and other pre-attentive attributes (eg. size, shape; see section 7) should be used deliberately to draw attention to particular aspects of the data in a way that isn't duplicative of other aesthetics, unlike in Figure 18. For example, the message of Figure 19 is about Victoria's unemployment rate compared to other states, so Victoria is highlighted. This is an appropriate use of colour. The other bars are de-emphasised using grey – they are there to give context to Victoria's number.

Figure 19: Victoria has the country's highest unemployment rate

Unemployment rate by state, Australia (July 2024)



Note: seasonally adjusted.
Source: ABS Labour Force

If you are generating multiple reports that will sit alongside each other – such as in a report or presentation **– you should use consistent colours for categories across charts**. For example, if you have many charts comparing unemployment rates across age groups, try to ensure that the colour for young people remains the same across charts. This helps readers quickly understand your visualisation.

8. OCE style: use text appropriately

Text is a vitally important part of any data visualisation, perhaps the most important. Use text carefully and deliberately, labelling and annotating your chart to illustrate your message. Format text consistently, in the OCE style. Integrate text with your chart, directly labelling and annotating, to help your readers understand your point.

DEWR's standard in-house font is Aptos**. You must use Aptos** in all visualisations and publications, unless there is a compelling reason why a non-standard font should be used.[16]

The default for text in DEWR OCE visualisation is **standard Aptos, 11pt** for visualisation in documents and **standard Aptos, 18pt** for presentations. Presentations should have larger text, as they are intended to be viewed from a distance.

Ensure your text is large enough to be clearly legible to your readers or audience. The appropriate size differs by context. The defaults set out in Table 2 are minima – err on the side of making your text bigger, not smaller.

Annotate your charts. Your annotations should support and reinforce the message of your chart, drawing the attention of your audience to the most important aspects of the data.

Table 2: OCE minimum sizes for text in visualisations

Element	Font	Colour	Size in documents	Size in presentations
Chart title	**Aptos Bold**	#404246	11 or 12pt	36
Chart subtitle	Aptos	#404246	11pt	18pt
Notes and sources	*Aptos Italic*	#404246	8pt	9pt
Axis labels, titles and legends	Aptos	#404246	11pt	18pt
Annotations	Aptos	#A4A7A9 or data colour	11pt (smaller in rare cases)	18pt (smaller in rare cases)
Direct labels	Aptos	Data colour	11pt	18pt

Note: these minimum sizes should be used in most cases. With justification, different sizes can be used. Err on the side of bigger, not smaller text.

[16] Aptos replaced Calibri as DEWR's default for in-house documents in mid-2024. If Aptos is unavailable in the visualisation software you're using, you can use an alternative sans serif font such as Helvetica or Arial.

Guidelines for text in charts

Text in OCE charts should follow these guidelines:

- X-axes should sometimes, but not always, have a short title. If the x-axis units are obvious, such as a date-time axis, an axis title is not necessary.
- Y-axes should generally not have a title. The chart's subtitle generally describes the y-axis units. An exception to this is scatterplots.
- Don't duplicate units (eg. "millions" or "per cent") – generally either put these in the axis title or chart subtitle, not both.
- If you deviate from the OCE default sizes in Table 2, generally err on the side of making sizes bigger, not smaller, than the defaults.
- Use **bold** sparingly to highlight important text.

Use annotations to convey a message

Annotations can help convey your main message, reinforcing the active title and connecting it to the data. In some cases, your annotations might be simple, such as showing the data values from some relevant periods.

For example, Figure 20 has a few simple annotations showing the latest unemployment rate and the average over the 20 years prior to COVID. These reflect and reinforce the chart's active title. The annotations help the reader by reinforcing the chart's active title and by answering questions they might have (like "what exactly was the pre-COVID average unemployment rate?"). The annotations sharpen a simple line chart, emphasising and reinforcing the message.

Annotations are particularly important for less familiar chart types. Even something like a scatterplot, which is beloved by economists and data analysts, can be confusing and alienating to a non-technical audience. Thoughtful annotations can help.

Figure 20: Unemployment remains below its pre-COVID average

Unemployment rate, Australia (July 2024), per cent

Note: seasonally adjusted. Monthly data since February 1978, quarterly before.
Source: ABS Labour Force and ABS Labour Force, Historical.

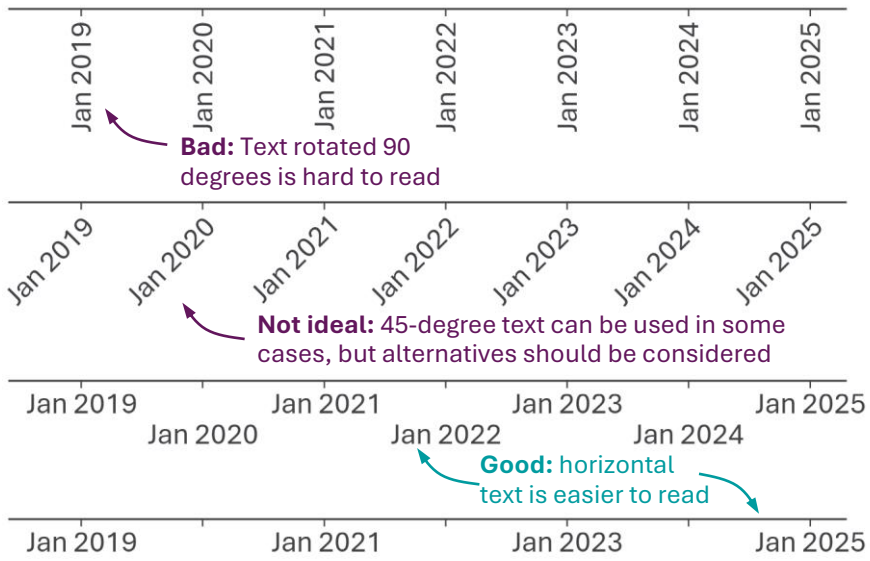
Consider the use of alt text to improve accessibility

Some readers with low vision rely on textual description of your charts. Rather than fully describing your chart in the body text of your document, **consider using alt text to describe what your chart shows** if your audience may include people with vision impairment.

Horizontal text: protect your neck

Text should generally be horizontal. Figure 21 shows two examples of rotated text – one at 45 degrees and the other at 90. **Do not rotate text 90 degrees** – it's too hard to read. Generally avoid rotating text 45-degrees for the same reason. In some circumstances, text may be rotated 45 degrees if alternatives (eg. a vertical bar chart) aren't viable.

Figure 21: 90-degree text is hard to read and should be avoided



Directly label your series

An over-arching goal for OCE charts is to make things easy on the reader. Reduce any frictions that might inhibit them getting the message from your chart. One way to do this is to **integrate text into your chart.** Directly label series where possible, rather than using a legend. Never do both.

Figure 22 and Figure 23 show the same data. The first chart has a legend, while the second directly labels the series. By integrating text with the chart, Figure 23 makes things easier on the reader. This chart also uses more intuitive, non-technical labels ("people employed in highest skill occupations") than the jargon-y descriptions in Figure 22's legend ("Skill Level 1 Occupations"). The text in the direct labels is coloured to match the corresponding series, with bold used sparingly to highlight the most important words. Variations on this technique, including labelling the end of each series, should be used where possible.

Figure 22: Charts with legends can be harder to read…

Number of people employed in occupations in skill level 1 and skill level 5, 1995-2005 (thousands)

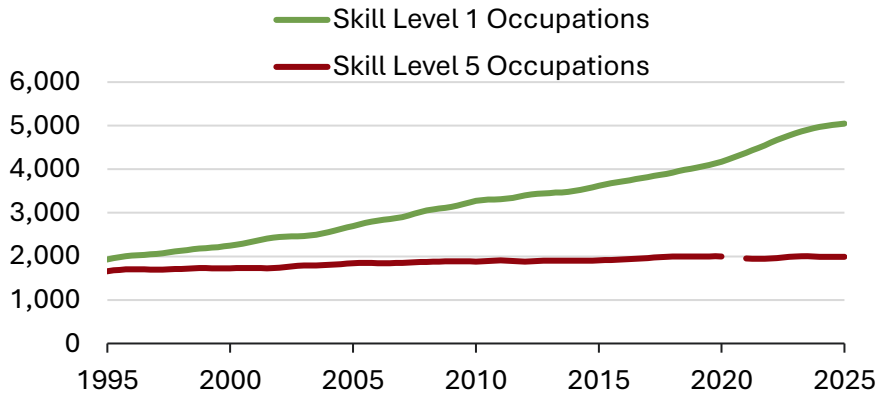


Figure 23: …than directly labelled charts

Number of people employed in the highest skill (level 1) and lowest skill (level 5) occupations, 1995-2005 (thousands)

Note: trended by JSA.
Source: ABS Labour Force, detailed.

9. OCE style: clean and minimal

OCE's clean minimal style reflects our focus on conveying a clear message and making it easy for the reader to understand this message. Using a consistent, professional style builds OCE's brand.

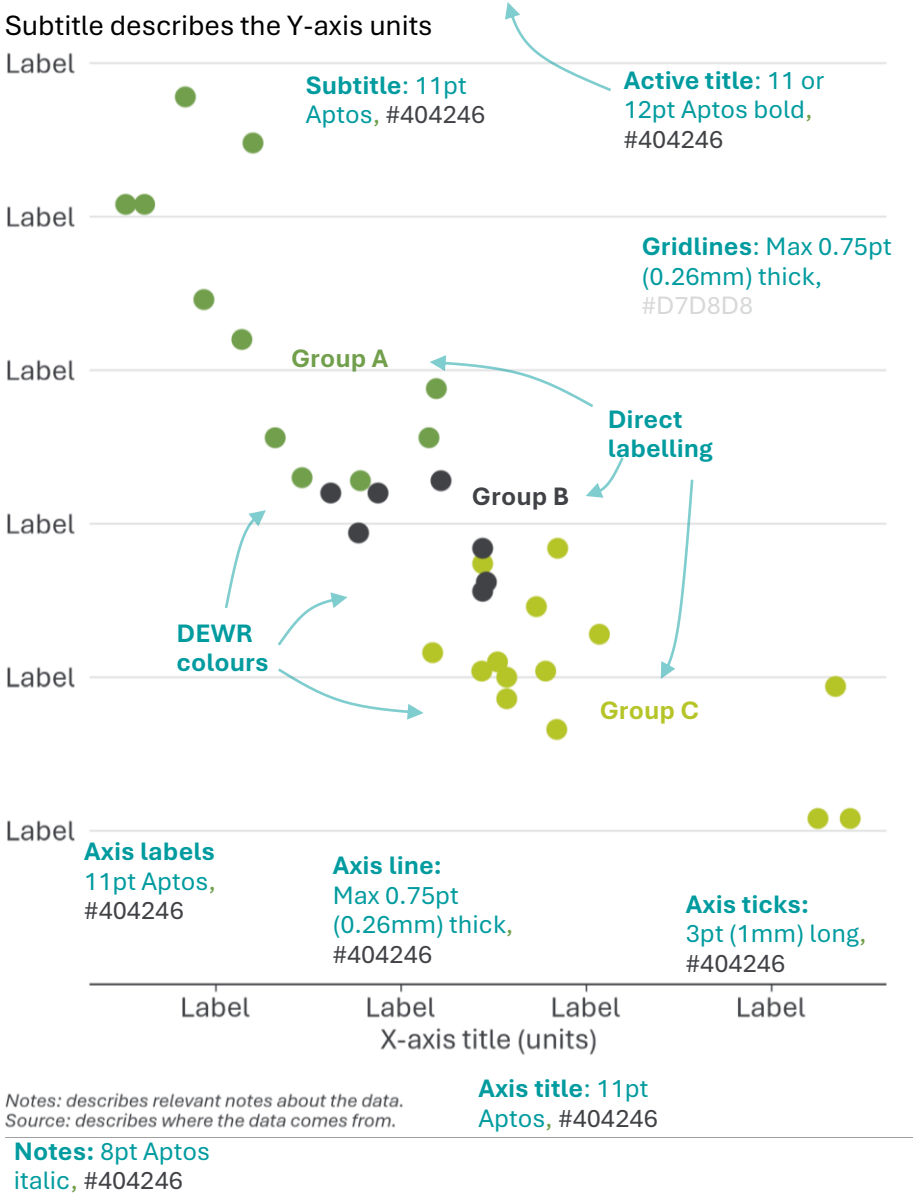
OCEans should eliminate or minimise visual elements ('chart junk') that do not help readers understand the data or the chart's key message. This is sometimes described as maximising the 'data:ink' ratio by stripping away non-data elements. This aggressively minimalist approach can be taken too far, but OCE's style does sit towards the minimalist, aesthetically clean end of the spectrum.

Some key stylistic elements of OCE charts are as follows:

- Gridlines are thin and light (max 0.75pt, #D7D8D8) or not present at all. If you have a bar chart in which each bar is directly labelled, for example, you don't also need axis lines and labels.
- We use colour deliberately, to convey information, not arbitrarily as decoration. Colours are draw from the DEWR palette. (See section 7, 'OCE style: use DEWR colours')
- We directly label rather than using a legend where possible, and never do both. (See section 8, 'OCE style: use text appropriately')
- OCE charts don't have borders, except in limited cases where we use them in sub-plots (facets, or small multiples).
- Axis lines and ticks are unobtrusive if present at all.
- We do not duplicate our axes.
- We don't use shading, patterns, or 3D effects.
- We avoid redundancy in visualisations – don't present the same information (like axis units) twice.

Figure 24 illustrates key elements of an OCE chart.

Figure 24: The elements of an OCE chart



The ‘little’ of OCE style

Style is built from many small decisions – what the data visualisation expert Andy Kirk calls the ‘little’ of data viz.[17]

- **Order series logically**. In a bar chart, or a similar chart type, always take care to present the order of the bars in a way that helps convey your message. Usually, this will involve ranking the bars from high to low (or vice versa). Don’t just order alphabetically or according to some other, non-data-driven factor. Data visualisation is about comparing numbers to one another – make it easy for your reader to make this comparison.
- **Don’t use unnecessary decimal places.** If all the axis labels on your chart are integers (eg. “10, 20, 30”), do not include a decimal point (“10.0, 20.0, 30.0”). The decimal point adds no additional information, but it does add clutter and visual complexity.
- **Use nicer date labels.** Microsoft products, by default, abbreviate months as “Mar-25”. There are good reasons for this in some contexts, like in a spreadsheet, but the hyphen is unnecessary in charts and adds clutter. Omit it and go with “Mar 25,” or if space permits, “March 2025”. Date axis labels can, and often should, break over two lines.

Chart size and shape

Specific OCE publication types (like LMI notes) have guidelines for default chart sizes. In general, an aspect ratio of 4:3 or 16:9 is usually appropriate for visualisations. Don’t go above 16:9 – your chart will be too wide, which can start to distort your message (for example, very wide charts tend to make variation over time in line charts appear less pronounced). Charts that are the same height and width (a 1:1 ratio) can work well in many contexts. The charts in this guide mostly (though not exclusively) have a 2:1 aspect ratio.

Never resize chart images. If your chart is inserted into your report or presentation as an image, you may be tempted to adjust its size. Never do this, even if you maintain the same aspect ratio. At best, the font sizes will be off; at worst, you will be left with a pixellated and stretched mess. Instead, resize the chart in the software used to create it, then re-export this resized version to your report.

Describing charts

OCE’s style is clean and minimal. This extends to the text in our reports that surrounds our charts. Don’t be tempted to describe and restate every aspect of the chart. The accompanying text should draw out the implications of your chart, not essentially duplicate it in text form. If you are tempted to write out the entirety of the chart’s message in the accompanying text, consider whether you need a chart at all.

[17] See Kirk’s blog post series on the ‘little of visualisation’ online: https://visualisingdata.com/2016/03/little-visualisation-design/

10. Take a bad chart and make it better

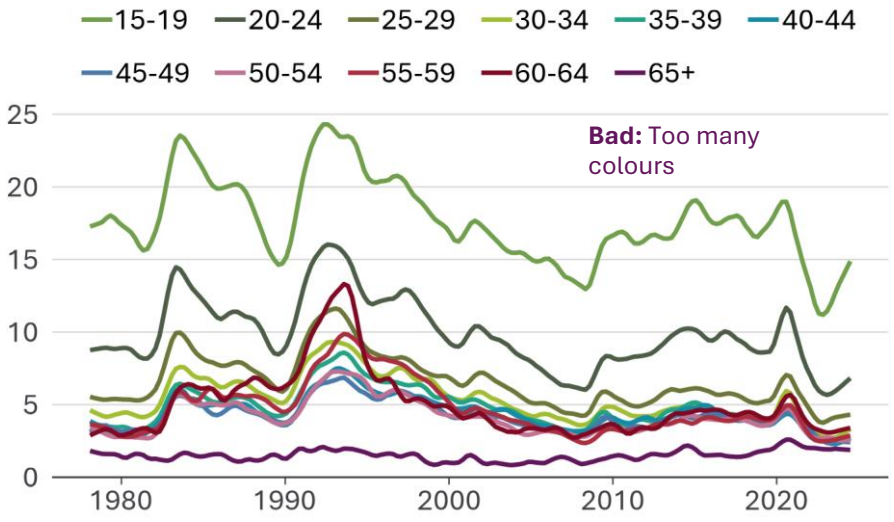
This section provides some examples of 'bad' charts and a range of ways to improve them.

Example 1: Spaghetti charts

A common type of bad chart is the 'spaghetti chart' – a line chart with too many series on it, like Figure 25. It has too many colours, it's too hard to see which colour corresponds to which series, and many of the series are so close to one another that they can't be distinguished. There's no hard and fast rule for how many colours is too many – but more than four is usually too many.

Figure 25: This is a bad chart with too many colours

Unemployment rate by age, Australia (per cent of labour force)

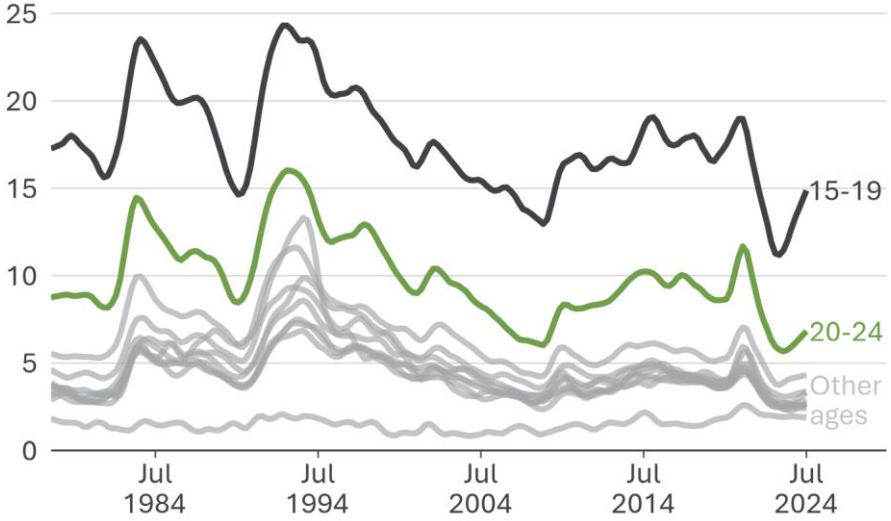


Note: trend estimates by DEWR.
Source: ABS Labour Force, detailed.

There are many ways this chart could be improved. Figure 26 provides one possible way to re-make this chart in an improved style. This chart focuses on the demographic group of interest – which in this case is young people. The other series are retained but are shown as semi-transparent grey lines that are present only to help contextualise the figures for young people. Figure 26 improves on Figure 25 in other ways – note the x-axis labels that end at the latest observation and provide the month, plus the direct labelling of the series in place of a legend.

Figure 26: Improve the chart by focusing on elements of interest

Unemployment rate by age, Australia (per cent of labour force)



Note: trend estimates by DEWR.
Source: ABS Labour Force, detailed.

Figure 27 shows another way to improve Figure 25. Instead of showing every detailed age group, we have aggregated these into three broad groups. This is a common way to improve a chart – by stripping out detail. Do not include extraneous detail for the sake of it – visualise the data that clearly makes your main point.

Figure 27: Improve the chart by aggregating groups

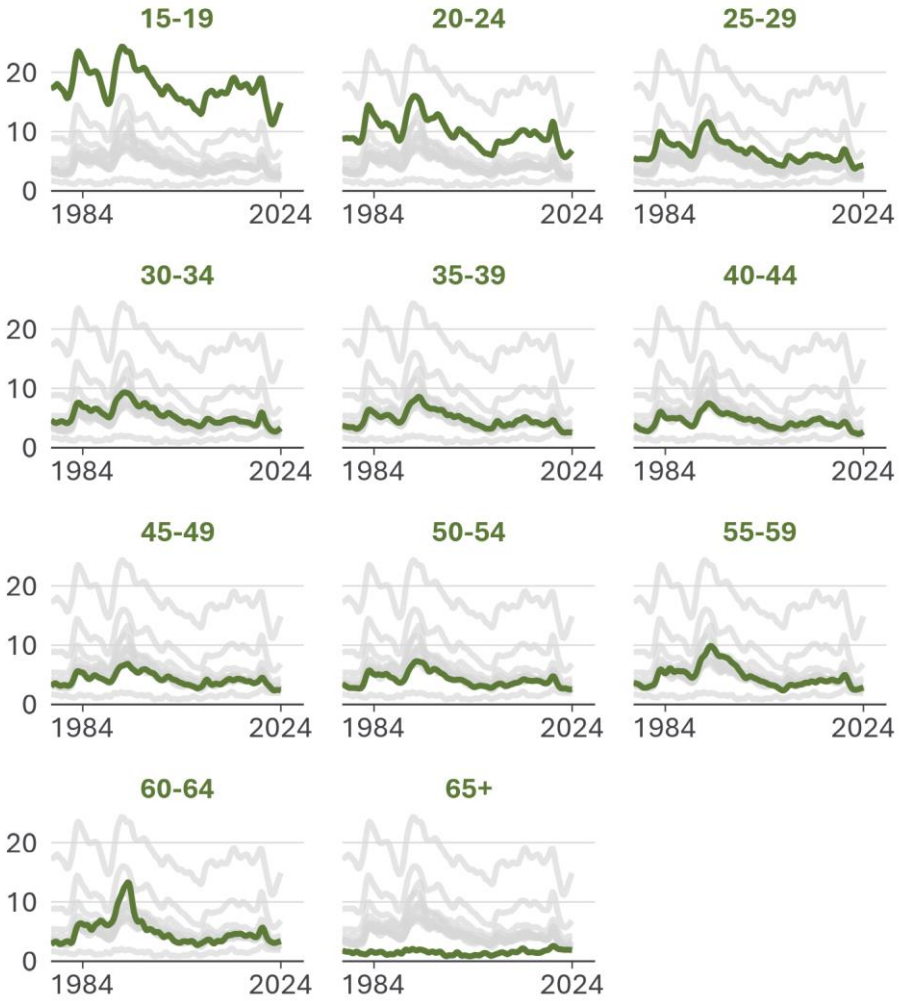
Unemployment rate by age group, Australia (per cent of labour force)

Note: trend estimates by DEWR.
Source: ABS Labour Force, detailed.

A final potential way to improve on Figure 25 is to use many small subcharts, sometimes known as facets or 'small multiples' (Figure 28). This may be appropriate if it is important to show each detailed group, but you don't want to show a spaghetti chart like Figure 25. Here, each group is broken out onto its own panel or subplot, with the other series shown as semi-transparent grey lines for context.

Figure 28: Improve the chart by using small multiples

Unemployment rate by age, Australia (per cent of labour force)



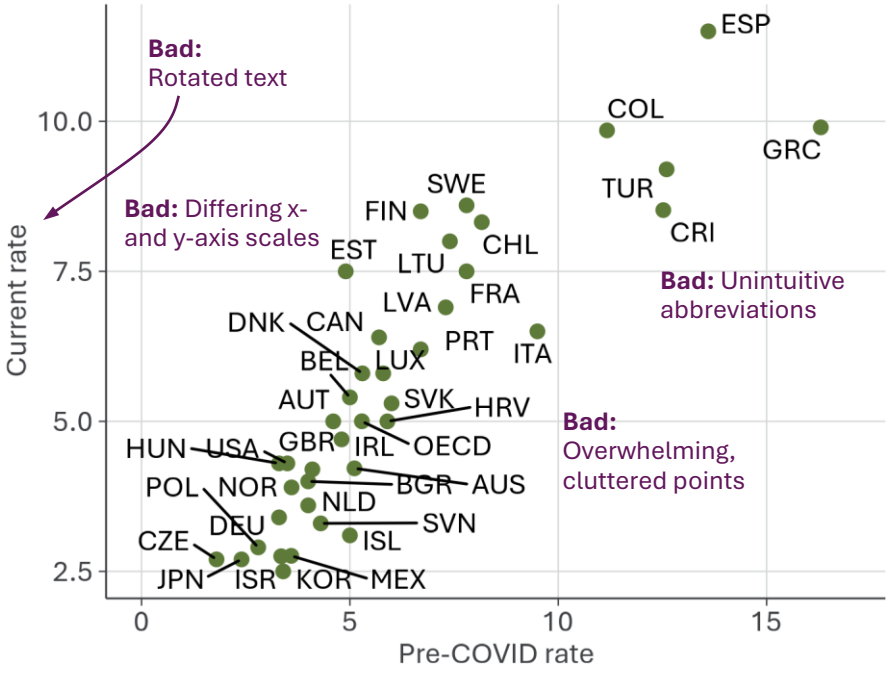
Note: trend estimates by DEWR.
Source: ABS Labour Force, detailed.

Example 2: Clutterplots

A common type of bad chart is the 'clutterplot'. A clutterplot is a messy, overwhelming scatterplot that usually features many points cluttered together and no obvious point. Figure 29 is an example. Let's say that in this case the chart's message is something like "unemployment in most OECD countries is around its pre-COVID level". Figure 29 does not make this point well.

Figure 29: Bad clutterplot is overwhelming and confusing

Unemployment rate, mid 2024 and Feb 2020 (per cent of labour force)

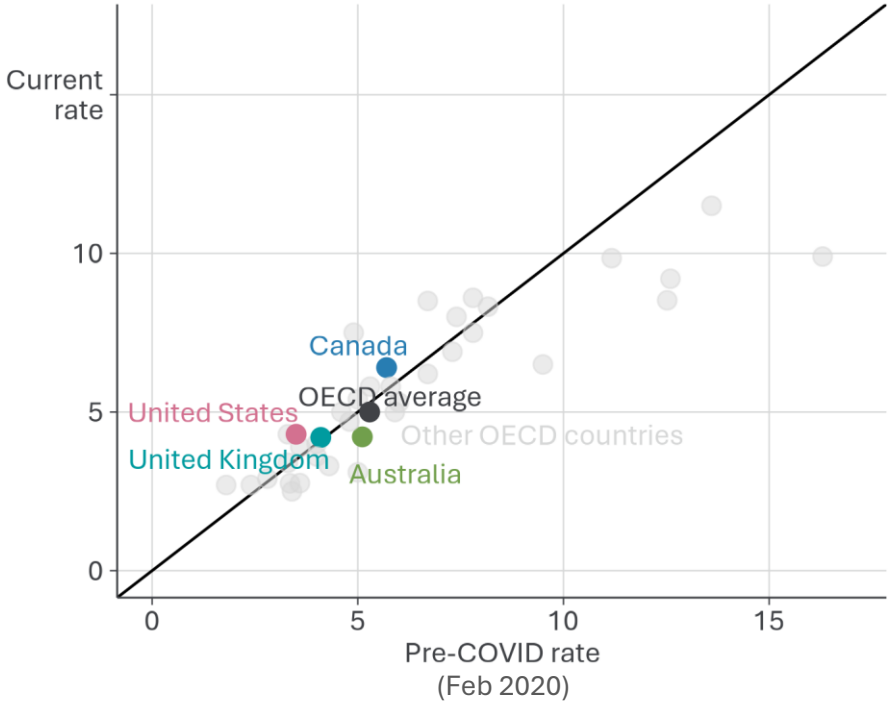


Note: Current dates are May 2024 for CRI and GBR; June 2024 for CHL, MEX, and TUR; and July 2024 for AUS, AUT, BEL, BGR, CAN, COL, CZE, DEU, DNK, ESP, EST, FIN, FRA, GRC, HRV, HUN, IRL, ISL, ISR, ITA, JPN, KOR, LTU, LUX, LVA, NLD, NOR, OECD, POL, PRT, SVK, SVN, SWE, and USA
Source: OECD Data Explorer

Figure 30 improves on the clutterplot. Key points, including Australia and the OECD average, are emphasised, while other OECD countries are de-emphasised. The inclusion of a 45-degree line, and the use of common x- and y-axes, makes it easier to see that most OECD countries are around their pre-COVID unemployment rates. The chart is also improved by using switching from abbreviations to full country names and not rotating text.

Figure 30: De-emphasise some data to make your point

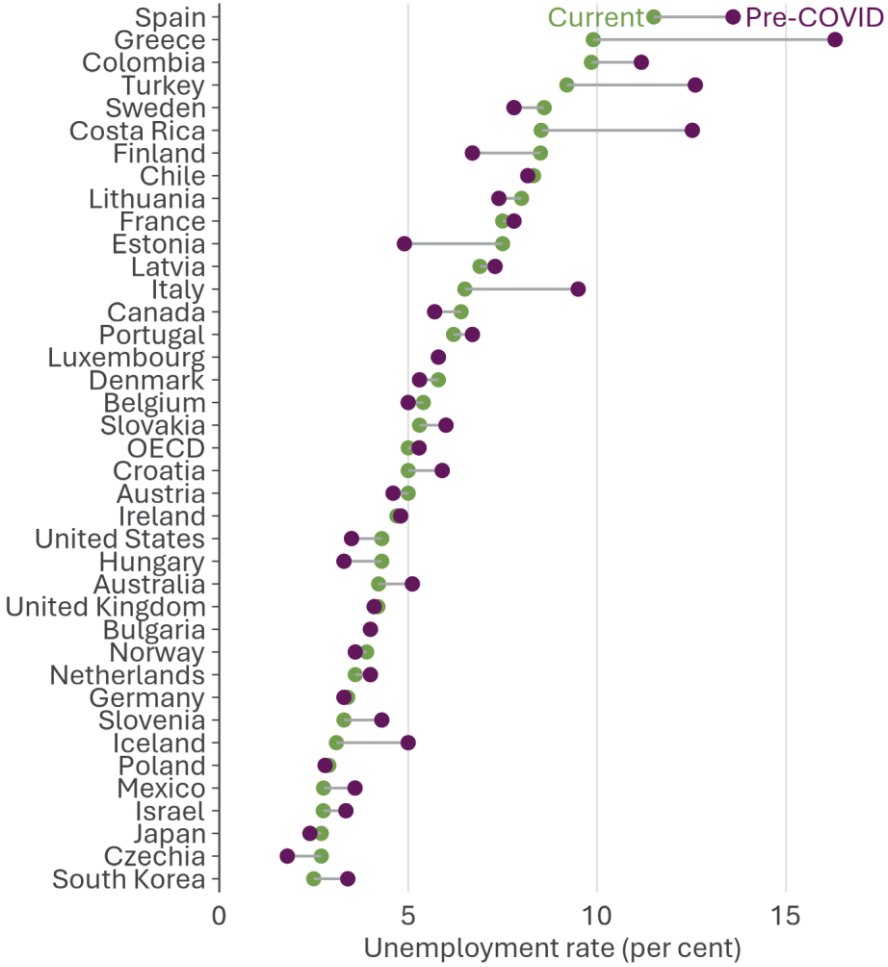
Unemployment rate, mid 2024 and Feb 2020 (per cent of labour force)



Note: Current dates are May 2024 for CRI and GBR; June 2024 for CHL, MEX, and TUR; and July 2024 for AUS, AUT, BEL, BGR, CAN, COL, CZE, DEU, DNK, ESP, EST, FIN, FRA, GRC, HRV, HUN, IRL, ISL, ISR, ITA, JPN, KOR, LTU, LUX, LVA, NLD, NOR, OECD, POL, PRT, SVK, SVN, SWE, and USA
Source: OECD Data Explorer

Figure 31: A dot plot might be a better choice

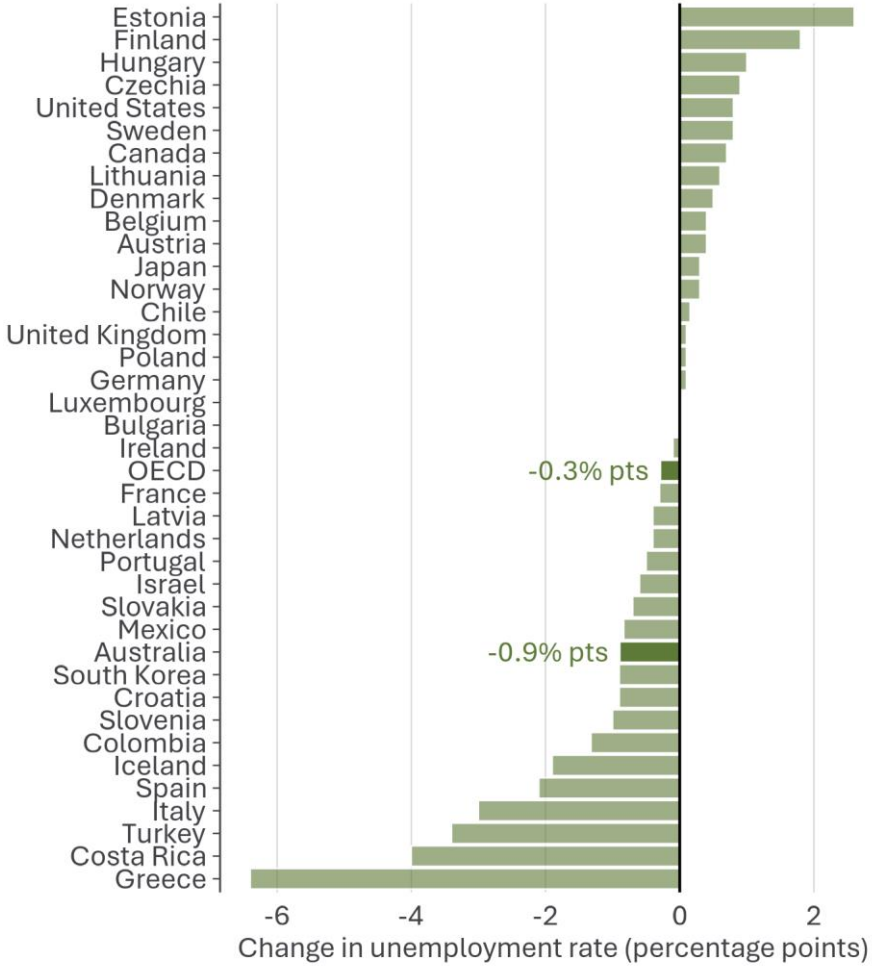
Unemployment rate, mid 2024 and Feb 2020 (per cent of labour force)



Note: Current dates are May 2024 for CRI and GBR; June 2024 for CHL, MEX, and TUR; and July 2024 for AUS, AUT, BEL, BGR, CAN, COL, CZE, DEU, DNK, ESP, EST, FIN, FRA, GRC, HRV, HUN, IRL, ISL, ISR, ITA, JPN, KOR, LTU, LUX, LVA, NLD, NOR, OECD, POL, PRT, SVK, SVN, SWE, and USA
Source: OECD Data Explorer

Figure 32: A simple bar chart tells the story clearest

Change in unemployment rate, Feb 2020-mid '24 (percentage points)



Note: Current dates are May 2024 for CRI and GBR; June 2024 for CHL, MEX, and TUR; and July 2024 for AUS, AUT, BEL, BGR, CAN, COL, CZE, DEU, DNK, ESP, EST, FIN, FRA, GRC, HRV, HUN, IRL, ISL, ISR, ITA, JPN, KOR, LTU, LUX, LVA, NLD, NOR, OECD, POL, PRT, SVK, SVN, SWE, and USA
Source: OECD Data Explorer

Although Figure 30 is a clear improvement on the clutterplot (Figure 29), it still may not be the best choice. If the message being conveyed is that

most OECD countries are around their pre-COVID unemployment rates, that message can be more clearly made with another chart type.

For example, Figure 31 is a dot plot showing the current and pre-COVID unemployment rates in OECD countries. In all but a handful of cases, like Greece, the current unemployment rate is around its pre-COVID level – in some slightly below where it was before COVID (eg. Australia) and in some slightly above (eg. the US). This conveys the same point as Figure 30 more clearly.

Figure 31 can be simplified further. If the focus is on the *change* in the unemployment rate, it may not be necessary to show the *level* of the rate at two different points in time. Simply plotting the change between the two periods is a clearer way to convey the central point. Figure 32 shows the difference between the current unemployment rate and the pre-COVID rate as a simple bar chart. The central point is made clearer by emphasising the OECD average; Australia is also emphasised to enable easy comparison.

Figure 30, Figure 31, and Figure 32 are all acceptable charts. None of them is 'wrong'. The appropriate chart will depend on the data being shown and the key message being conveyed. But a clutterplot like Figure 29 can always be improved. Identify what the chart's key message is, and then ensure that the chart type, design choices, and annotations all convey that key message as clearly as possible.

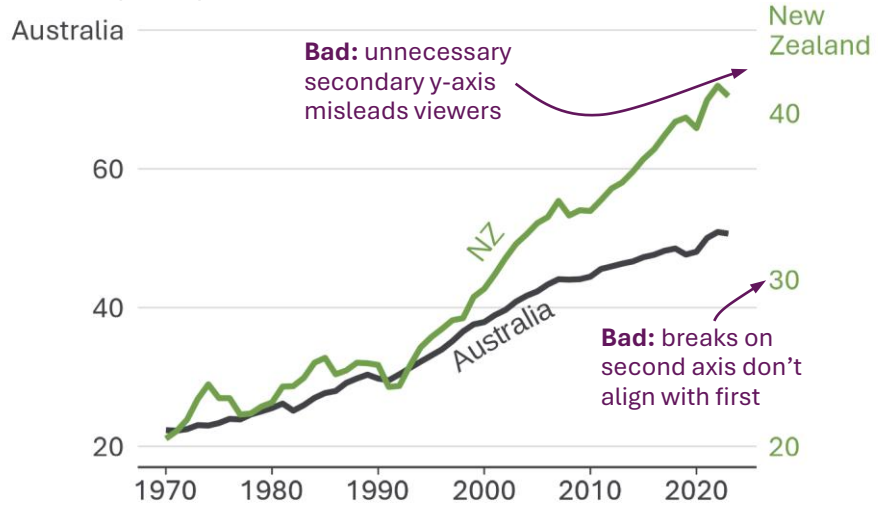
Example 3: Dual-axis line charts

As discussed in 'Be careful with dual y-axes' (section 4, p.8), charts with two vertical axes can easily mislead readers, even if this is not intended. Because of this potential to mislead, you should be very careful with dual-axis charts. Consider whether it's possible to visualise your data in a different way. If not, make sure you take care to ensure readers are not inadvertently misled by your chart.

Figure 33 is an example of a bad dual y-axis chart. Viewers could easily be misled into thinking NZ GDP per capita is higher than Australia or has risen faster, neither of which is true. The secondary axis is unnecessary and actively harmful.

Figure 33: This is a very bad, misleading dual y-axis chart

Real GDP per capita, **Australia** and **New Zealand**, '000s of USD



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, converted at purchasing power parity.
Source: OECD Data Explorer

There are many potential alternative ways to visualise the data in Figure 33. The simplest of these is just plotting the series on a common scale, as in Figure 34. This isn't always possible, but where it is (like in this example), it is usually the best way.

Another option is to show the data on separate subplots or facets, as in Figure 35. Here the line charts are shown above each other; they could

also be shown side by side. Because each series clearly has its own separate scale, it is not misleading like Figure 33.

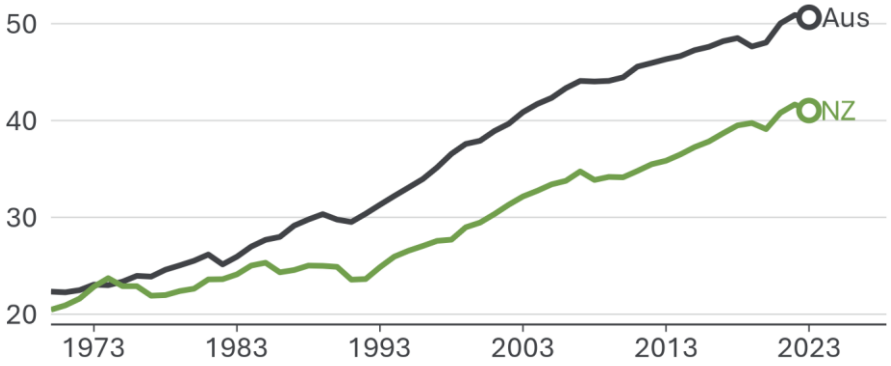
Converting the series to index numbers (Figure 36) is often a good alternative to a dual-axis chart. This is especially the case if the point of the chart is about the relative growth of the two series over some period of time. Take care in choosing the base year for the index.

In some cases, dividing one series by the other can work, as in Figure 37. The chart shows that in the early 1970s, Australian GDP per capita was around the same level as NZ's, before rising much faster in the 1980s and '90s. The ratio stabilised in the 2000s before falling a little in the 2010s. In some cases, subtracting one series from the other can also be illuminating.

For many dual y-axis charts, a scatterplot is the best replacement. Figure 38 and Figure 39 show regular and connected variations, respectively. In this case, the scatterplot is probably not the best option, but it is a good replacement for many dual y-axis charts where the point relates to the correlation between two series. For example, if you are visualising wages growth and labour underutilisation, a scatterplot is a good option that is clearer about the degree of association than a dual y-axis chart.

Figure 34: Plot the data on a common axis if you can

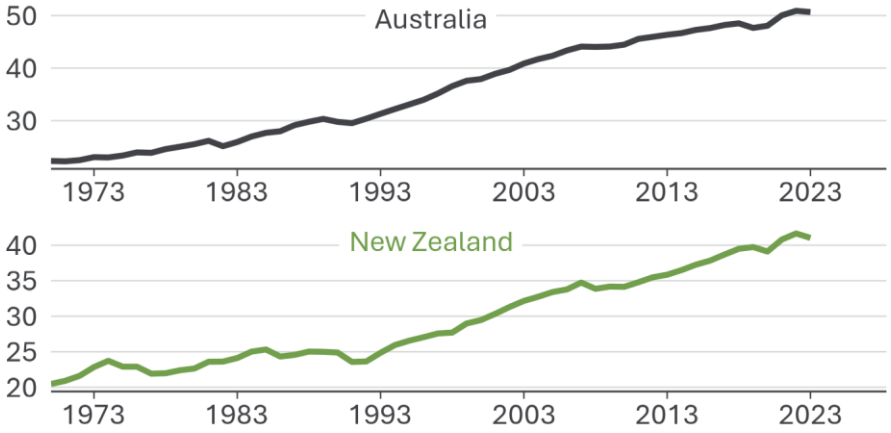
Real GDP per capita, **Australia** and **New Zealand**, '000s of USD



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, converted at purchasing power parity.
Source: OECD Data Explorer

Figure 35: Faceted plots can be a good alternative

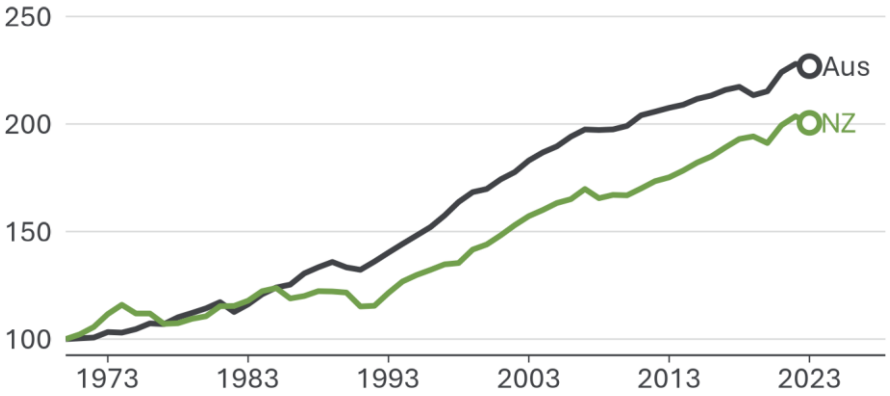
Real GDP per capita, **Australia** and **New Zealand**, '000s of USD



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, converted at purchasing power parity.
Source: OECD Data Explorer

Figure 36: Try converting the series to index numbers

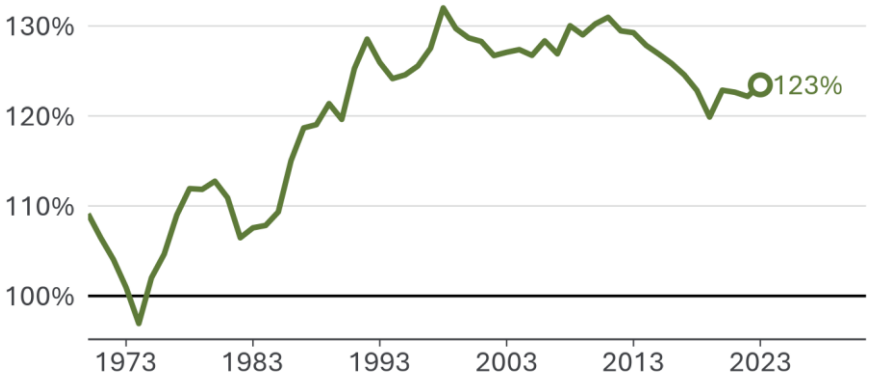
Real GDP per capita, **Australia** and **New Zealand** (Index: 1970=100)



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, converted at purchasing power parity.
Source: OECD Data Explorer

Figure 37: Dividing one series by the other can work well

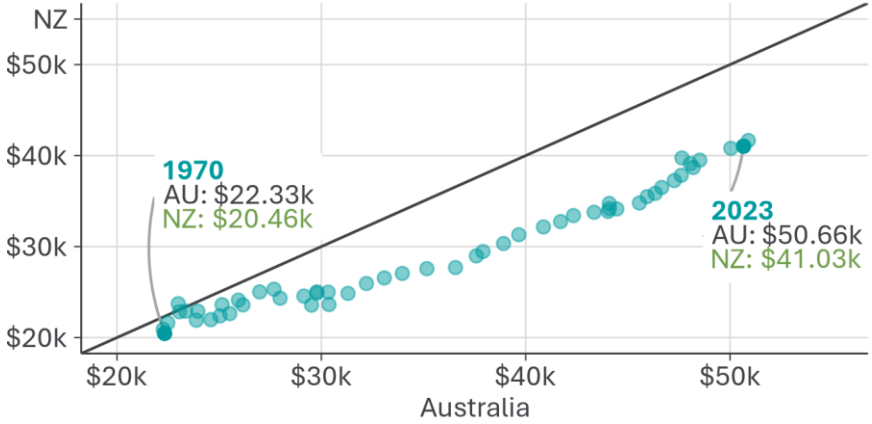
Australian GDP per capita as a percentage of New Zealand's



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, converted at purchasing power parity.
Source: OECD Data Explorer

Figure 38: A scatterplot is often a good option

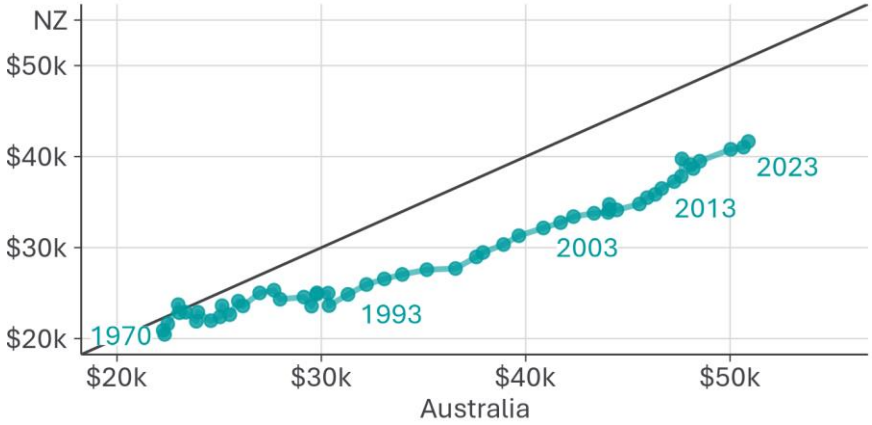
Real GDP per capita, Australia and New Zealand, '000s of USD



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, at PPP.
Source: OECD Data Explorer

Figure 39: A connected scatterplot sometimes works

Real GDP per capita, Australia and New Zealand, '000s of USD



Note: figures are adjusted for inflation and expressed in constant 2015 US dollar terms, at PPP.
Source: OECD Data Explorer

11. Other resources

There is a large range of excellent materials available to help you learn to make excellent, high-impact data visualisations.

Some great resources are:

- Clause Wilke's book *Fundamentals of Data Visualization,* which is also available free online: https://clauswilke.com/dataviz/. Like this guide, it is a "tool-agnostic" guide to principles of visualisation. **All OCEans should read Chapter 29, 'Telling a Story and Making a Point.'**
- Kieran Healy's book *Data Visualization: A Practical Introduction,* which is also available free online: https://socviz.co/. Healy's book covers general principles, but also provides clear, worked examples to make ggplot2 charts in R.
- Jon Schwabish's book *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks:* https://cup.columbia.edu/book/better-data-visualizations/9780231193115/
- Although Jon Schwabish's book (above) isn't available free online, his 2014 article for the *Journal of Economic Perspectives* is available free and is an excellent introduction to key principles: https://pubs.aeaweb.org/doi/pdfplus/10.1257/jep.28.1.209.
- The FT's Visual Vocabulary provides a great overview of some different types of charts that can be used for specific purposes: https://journalismcourses.org/wp-content/uploads/2020/07/Visual-vocabulary.pdf.

12. Issues not covered

These guidelines, long though they may seem, do not come close to covering the field of data visualisation issues relevant to our work in OCE. Future versions of the guidelines may expand to cover other issues, including:

- Which chart type to choose[18]
- Interactive dashboards
- Tables
- Animated charts
- Continuous and diverging colour scales

[18] Consult the FT's Visual Vocabulary: https://journalismcourses.org/wp-content/uploads/2020/07/Visual-vocabulary.pdf